Bailey Hall evacuation modelling

Mathematical specification, assumptions and verification

Abstract

This document develops the model from flow conservation and travel times to heterogeneous agents, spatial interactions, imperfect guidance and the A-Lite+ primary model, with physical installations, local information and explicit instruction execution. The original A/B/C/Lite engines remain reference models. Equations describe the implemented rules; citations justify modelling mechanisms without treating unmeasured coefficients as Bailey-specific evidence. The audience remains 1,329 people, including three wheelchair users. The quantity of interest is the mean, over independent audiences, of the last person's exterior-boundary crossing time. The companion experimental reports distinguish numerical corrections, structural simplifications and parameter calibration.

Study	Bailey Hall, full-house evacuation
Audience	1,329 occupants, including three wheelchair users
Base revision	Mateus, `3e4bcf0`, 8 November 2026, 04:20:42 EST
Report date	8 November 2026
Scope	Research model; assumptions and calibration limits stated explicitly

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From flow arithmetic to spatial people, adaptive guidance and reproducible comparisons

Technical specification continuing Mateus's **8 November 2026, 04:20:42 EST commit** `3e4bcf0`, implemented in `CMCM/mateus-0420`. The original A/B/C models, the A-Lite transport approximation and the final A-Lite+ guidance extension are distinguished throughout.

The problem remains a full house of **1,329 people**, including three wheelchair users. Within each compared model, the base geometry, discharge parameters, speed distributions, intrinsic response times and movement rules are preserved in the primary intervention study. Physical barriers may reduce usable width or close an existing passage as explicitly modelled. A-Lite's separate group-motion abstraction is stated in Section 19; A-Lite+ changes local group pace, perception and instruction execution as specified in Section 24, then freezes them before operational comparisons. The target is the mean time for the **last person to leave the building**, below 180 seconds.

The companion **RESULTS_AND_CHANGES.pdf** contains the completed experiment tables, selected configurations, uncertainty, rejected alternatives and changes from the 04:20 commit. The equations here explain the implementation; a cited mechanism does not make its numerical parameters measured Bailey data. **A-Lite+ is the primary model for the final comparative planning study (Section 24)**, subject to the scope and calibration limits in Sections 22–24. It retains the detailed walkway graph and individual behavior while representing movement by finite queues. A and B remain checks on sensitivity to model structure, and C is a capacity benchmark.

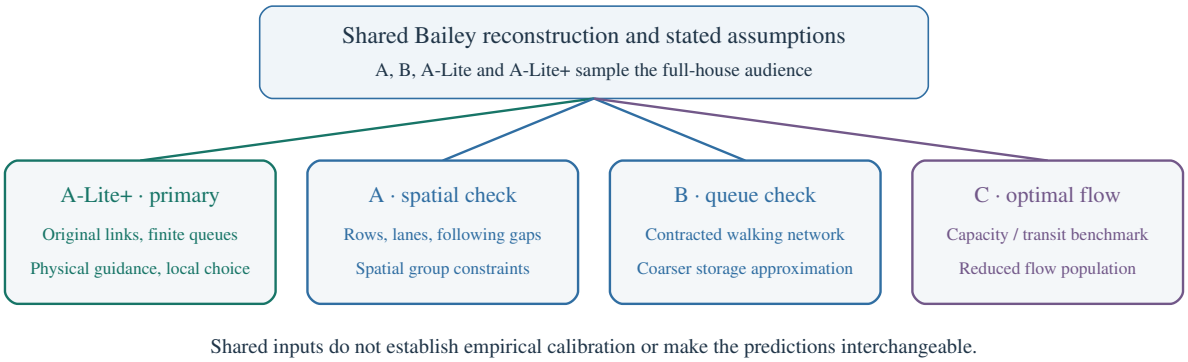


Figure 1: A-Lite+ is the primary detailed-network queue model; A, B and original Lite provide structural comparisons, while C is an ideal flow benchmark.

Finalization scope (8 November 2026). The user requested closure using completed work. The companion report contains 2,430 completed experiment replications. The retuned twelve-manager policy at 1.4 m/s and a 0.0125 s step averages **184.45 s** over 100 audiences (95% CI **182.39–186.51**); the sub-180 mean is not demonstrated. Supplemental ablations, device-isolation tests, Q1/Q3 scenario batches, parameter uncertainty, two other 100-run fine-step replays and the 0.00625 s diagnostic were not completed. Obstruction tests are partial. Later sections describe both the implemented model and declared experimental designs; a declared test is not a completed result.

1 Notation and reading order

Read the flow and queue sections first, then the shared population and choice rules, then spatial Model A and interventions. The final sections introduce numerical verification, A-Lite and the targeted A-Lite+ extension. Read Section 24 after the shared foundations to see exactly which inherited rules it changes. The equations are definitions of the simulation, not fitted laws for Bailey. Numbered citations identify the evidence supporting a mechanism; a numerical coefficient remains an assumption unless stated otherwise.

Table 1: Notation and reading order.

Symbol	Meaning	Units
N, R, i, e, g	Audience size, replications, person, link and exit-group indices	Counts / indices
$t_i, T_r, \overline{T}$	Individual exterior crossing, last crossing in run r, mean across runs	s
L_e, w_e, r_i	Link length, width and body radius; r_i denotes release time only in (5)	m; release in s
$v_i, H, \Delta t$	Free speed, net following headway, simulation step	m/s, s, s
C_e, J_e, Q_e	Gate rate, passage rate, queue occupancy	persons/s, persons/s, persons
ρ, S_e, b	Occupancy density, storage, service credit	persons/m^2, persons, persons
a_j, d_j	Arrival and service-departure times of queue customer j	s
V_{ig}, ε_{ig}	Systematic utility and persistent random preference	Dimensionless

Symbols are local to their equations where the inherited notation overlaps: H is a time-layer horizon in (4), and an audience count in (16c); b denotes boundary width in (2) and service credit in (7). The sets $\delta^-(v)$ and $\delta^+(v)$ are incoming and outgoing links at node v. $\mathcal{P}_i$ contains routes accessible to person i. In A-Lite, g_e is the existing geometric area correction, and l_e is the number of lanes.

Worked flow calculation. A representative person needs centre spacing $d + vH$: body diameter d plus the distance vH walked during the following time H. One person passes a point every $(d+vH)/v$ seconds. Taking its reciprocal gives $J = v/(d + vH)$. For d=0.44 m, v=1.2 m/s and H=0.7 s, spacing is 1.28 m and discharge is 0.9375 persons/s per lane. This derives the rate used by the queue approximation; it does not assert all actual people have that speed.

Worked queue calculation. At a 2 persons/s gate, three arrivals at 0, 0.2 and 0.3 s leave service at 0.5, 1.0 and 1.5 s under equation (6), starting from an empty gate at time zero. Each service takes 0.5 s; overlapping arrivals wait. In a building network, downstream storage can postpone admission further. Travel and service therefore cannot be replaced by simply dividing total occupancy by summed door rates.

How the simulation advances. Sample one audience and its persistent traits; update awareness, preparation and group readiness; observe guidance and choose eligible routes; apply movement or travel clocks; admit people subject to service and downstream space; record the first exterior crossing; repeat until all are outside or the failure horizon is reached. Model C optimizes a flow schedule rather than executing these individual choices. A-Lite changes the movement stage while retaining shared population and guidance logic.

2 What is measured, and what remains fixed

For replication r, let t_i be person i's first crossing of an exterior building boundary. An assistance point outside the building counts as outside; it does not mean assisted travel to street level has finished.

$$T_r = \max_{1 \le i \le N} t_i, \qquad \overline{T} = \frac{1}{R} \sum_{r=1}^{R} T_r$$
$$\widehat{P}_{180} = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}\{\text{all } N \text{ outside and } T_r \le 180\} \tag{1}$$

$\overline{T}$ is an average of **last-person times** across audiences. The average of individual exit times and the 95th percentile of individual exits answer different questions. A run with anyone unresolved has no finite full-house clearance time; it is a target failure, never silently removed from a claimed successful mean.

Table 2: What is measured, and what remains fixed.

Quantity	Meaning and fixed scope
Audience	527 orchestra + 486 mezzanine + 316 balcony = 1,329
Accessibility	Three wheelchair users remain; accessible routing and assistance retained
Movement	Same row speeds, body sizes, headway, stairs, door rates and groups
Interventions	Staff posts/count, exit messages, pre-announcement, LEDs and barriers
Primary exclusions	Wider aisles, faster people, faster intrinsic response, perfect compliance, removing group waiting

Mateus's `package-180` also applies a voice alarm and universal agreement to meet outside. It is reproduced as a separate inherited package, not counted as a guidance-and-barrier-only result. The 900-second horizon is a failure-detection limit, not a way to censor a run at 180 seconds.

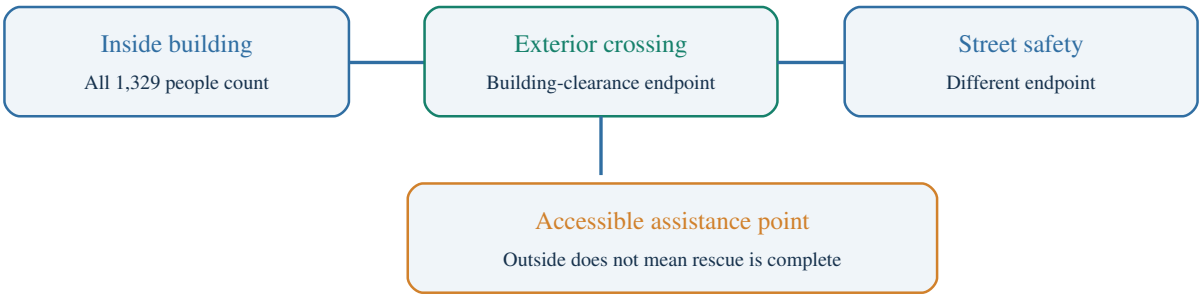


Figure 2: Building departure and street safety are separate endpoints.

3 The model ladder and the building graph

Start with one exit and a rate. Add connected passages, then travel times and finite queues. Add heterogeneous people, their knowledge and preferences, spatial interactions and finally interventions. These are explanatory layers, not eight separate calibrated simulators.

Table 3: The model ladder and the building graph.

Layer	Adds	Implemented in
V0: rate	Persons / persons per second	Arithmetic check
V1: network	Route connectivity and bottlenecks	C static flow

Continued on next page

Table 3 continued

Layer	Adds	Implemented in
V2: time	Travel, release and storage	C time-expanded flow
V3: people in queues	Individual arrival, FIFO service	B
V4: response	Delays, knowledge, groups	Shared A/B population
V5: choice	Travel, queues, habit, guidance	A/B routing
V6: space	Rows, gaps, lanes, counterflow	A
V7: information	Staff, signs, beliefs and redirection	A/B, partial policy A only

The detailed graph G=(V,E) has junctions and row ends as nodes; walkable links retain width w_e in metres, length L_e in metres, floor, slope, accessibility and direction. Seven exit groups collect individual doors. The network contracts chains while preserving door, stair and barrier-slot identities; its dimensions are generated from the same JSON geometry.

The geometry is a reconstruction, not a surveyed digital twin. Near symmetry is encoded, but actual seating, paths, queues and sampled groups need not be symmetric. All simulations use the unchanged geometry file; its SHA-256 is stored with the results. A symmetric plan is a candidate to test, not an assumption that paired runs must be identical.

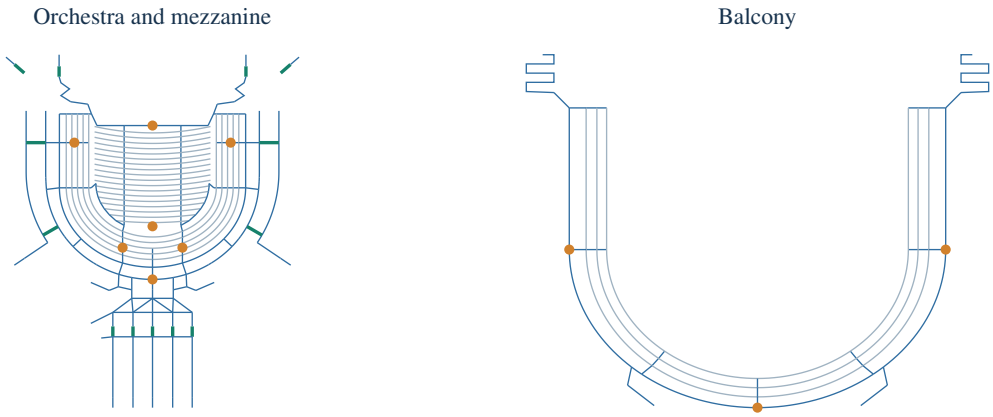


Figure 3: The unchanged main and balcony graph, with existing staff posts.

4 V0–V1: rates, capacities and bottlenecks

An exit with sustainable rate C persons/s needs roughly N/C seconds of service after people reach it. The required whole-run average discharge is 1329/180 = **7.383 persons/s**. Comparing this number with summed exit capacities alone ignores how people reach those exits.

$$\begin{aligned} C_e &= q_e w_{\text{eff},e} f_{\text{luggage}} f_{\text{env}} g(u,\pi) \\ w_{\text{eff}} &= \max(0.4, w - 2b) \\ g(u,\pi) &= (1+au)\left[1 - d\max\left(0, \frac{\pi-\theta}{1-\theta}\right)^2\right] \end{aligned} \tag{2}$$

q has units persons/(m·s); effective width removes boundary layers. The default door rate is 1.3, stair rate 1.0; boundary layers are 0.15 m at doors and 0.09 m at stairs. Luggage factor is 0.97. u is configured

urgency, π is urgency times mean queue impatience. The factors a=0.15, d=0.45 and θ=0.35 define Mateus's faster-is-faster / faster-is-slower hypothesis. For stairs with a lane barrier, effective width is instead lanes × max(0.3, width/lanes − 2b), applying boundary loss to each channel. No obstacle adds a positive multiplier to this equation.

For a directed network flow f_e, capacities and conservation give:

$$0 \le f_e \le C_e, \qquad \sum_{e \in \delta^-(v)} f_e = \sum_{e \in \delta^+(v)} f_e$$
$$F^* = \max\{\text{source-to-exit flow}\} = \min\{\text{cut capacity}\} \tag{3}$$

A cut is a set of links that every source-to-exit flow must cross. The max-flow/min-cut result [R1] identifies structural restrictions. N/F* is a capacity benchmark, not a prediction of finite-population completion. Supply is uneven, people are not all ready, walking takes time, and congestion begins upstream. `Static Flow.ts` connects unlimited-rate super-sources to row-end supplies and exterior sinks; it does not confuse population counts with rates.

For walkways, representative lane flow is J=v/(d+vH), where d is two mean body radii and H is time headway. With v=1.2 m/s, d=0.44 m and H=0.7 s, J≈0.938 persons/s per lane. B uses this same representative capacity. A additionally resolves individual gaps, speeds and groups. The unchanged lane count is max(1, floor(width/0.6 + 0.2)): two lanes at 1.1 m width, one at 1.0 m. These values are preserved, not fitted to the target.

5 V2: the time-expanded benchmark, Model C

Dynamic flow [R2] makes a copy of each graph node at each discrete time kΔt. Movement connects an upstream node now to its downstream node after travel; waiting connects consecutive copies of the same node. Supplies enter when released. Exterior copies connect to a super-sink.

$$\begin{aligned} \tau_e &= \operatorname{round}\left(\frac{L_e}{v_{\text{rep}} f_{\text{stair}} \Delta t}\right) \\ c_{e,k} &= \lfloor C_e(k+1)\Delta t \rfloor - \lfloor C_e k \Delta t \rfloor \\ S_v &= \rho_{\max} A_v, \qquad T^* = \min\{H\Delta t : \text{maxflow}(H) = N_{\text{flow}}\} \end{aligned} \tag{4}$$

$c_{e,k}$ counts people per time slice and conserves fractional capacity cumulatively. S_v is holding storage in persons, using the assigned node area A_v. Movement capacities, holdover capacities and nonnegative flows constrain the solution. Extending the graph one layer at a time yields the cumulative earliest-arrival curve. Default Δt is one second; rounded short travel times can become zero, so discretization error is reported.

The immediate case releases everyone at zero plus row walking. The intrinsic case releases them after sampled recognition, response and preparation. A conditional case can use actual starts from A or B. **C's flow component excludes wheelchair users** and uses one representative speed; it is not itself a full-house, heterogeneous behavioral evacuation result.

An additional own-speed diagnostic computes:

$$B_{\text{ind}} = \max_i \left\lceil r_i + \min_{p \in \mathcal{P}_i} \sum_{e \in p} \frac{L_e}{v_{i,e}} \right\rceil \tag{5}$$

r_i is the supplied departure time, with row walking added separately in code. Accessible links are required for wheelchair users. The correction in this continuation reports an infinite/unresolved diagnostic for

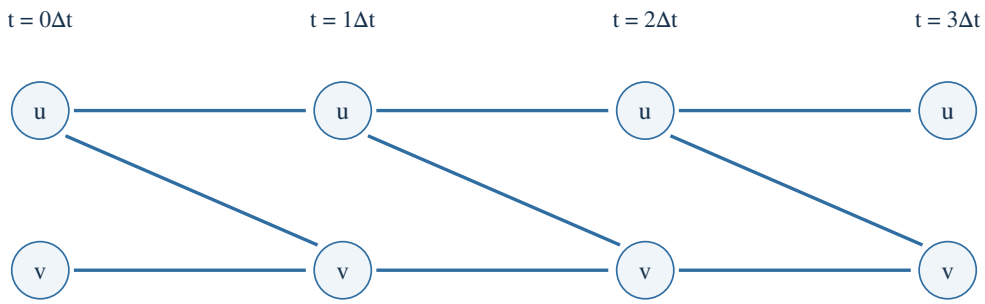


Figure 4: Movement arcs carry people forward in both space and time; holdovers represent waiting.

unknown starts or inaccessible paths instead of dropping those people. The intrinsic-release benchmark is conditional on those releases: staff and social effects can change actual starts. Nor is max(C, B_{in}d) a universal rigorous bound for every heterogeneous-speed spatial run.

6 V3: stochastic queues, Model B

Model B gives every person a state, a route, a queue entry time and a transit time. Each directed arc has a FIFO queue and a service-credit account. After admission, transit takes $L_e/(v_i\ f_{stair})$, using the slowest member's free speed for intact groups. At the receiving node, the person chooses or continues a path. Walking lanes are contracted; microscopic collisions are not simulated.

$$\begin{aligned} d_j &= \max(a_j, d_{j-1}) + \frac{1}{C} \\ Q(t+\Delta t) &= Q(t) + \text{arrivals} - \text{admissions} \end{aligned} \tag{6}$$

For a fixed-rate FIFO bottleneck, a_j and d_j are arrival and departure times in seconds. Equation (6) explains queue service; the engine approximates it with discrete tokens rather than scheduling every exact departure. Default B step is 0.25 s. Counterflow shares an edge's capacity according to directional demand, including waiting and in-transit people.

The inherited storage rule checks people waiting at the receiving node plus transit on the particular incoming arc. After 30 seconds of storage blockage it permits relief. **It is not a strict reservation of all incoming traffic**, and relief can exceed nominal storage. This optimistic/deadlock-relieving assumption is retained and disclosed; B is not evidence that spatial crowd interactions disappear. Its optional density-speed factor is off by default.

The continuation adds a separately switchable service correction:

$$\begin{aligned} b^* &= b_k + C_k \Delta t, \qquad n_k \le \lfloor b^* \rfloor \\ b_{k+1} &= \min(1, b^* - n_k) \end{aligned} \tag{7}$$

b is service credit in persons; n is actual admissions allowed by both service and space. Bounding residual credit **after** service preserves fractions while preventing idle hoarding. Mateus's original `min(1,b+CΔt)` before service discards fractions under saturation and caps any arc at one admission per step. The correction changes numerical integration, not C. Analytical tests verify rate conservation across time steps and preserve responder reductions. Default/absent correction retains the 04:20 behavior exactly.

7 V4: the audience, response and social groups

`createPopulation` is shared by A and B. Named pseudorandom streams separate the audience from later routing, hearing and adaptive traits. The same seed produces the same seats, speeds, response draws, group membership and compliance uniforms under primary interventions.

$$\begin{aligned} \tau &= \exp(\ln m + \sigma Z), \qquad Z \sim \mathcal{N}(0,1) \\ m_{\text{response},i} &= 14 L_{\text{level}} \exp\left[0.04 \min(d_{\text{aisle}}, 15) + 0.05 \frac{d_{\text{door}}}{10} - 0.15 I_{\text{familiar}}\right] \end{aligned} \tag{8}$$

m is a median in seconds; σ is the standard deviation on the log scale. Recognition median is 6 s, response base median 14 s, both σ=0.6. Preparation is a mixture: half the audience collects belongings with median 6 s and σ=0.6; the remainder has a short preparation phase with median 1.5 s and σ=0.3. Separate standard-normal draws are used for recognition, response and preparation. Theatre evidence supports lognormal response and position dependence [R5]; these particular coefficients are inherited assumptions, not Bailey measurements.

The behavioral states are unaware → aware/response → preparing → waiting → moving. Recognition and response clocks advance faster when seated neighbors are standing or moving; preparation advances at real time:

$$\begin{aligned} \frac{dc_i}{dt} &= 1 + 1.5\varphi_i, & p_{\text{aware}}(\text{step}) &= 0.4\varphi_i \Delta t \\ \tau'_{\text{response}} &= c_i + 0.5 \max(0, \tau_{\text{response}} - c_i) \end{aligned} \tag{9}$$

φ is the local moving share plus half the preparing/standing share. The awareness code uses the first-order hazard probability shown above; its continuous-time analogue is $1-\exp(-0.4\varphi\Delta t)$. The last expression applies after a first heard manager instruction. It changes realized response through the inherited mechanism; the raw response distribution is not resampled or reduced by this study.

Half the audience is grouped, approximately, in adjacent groups of 2–4. Members wait for readiness up to 90 s, follow a known leader exit, maintain cohesion and can break apart beyond 12 m. Nearby faster members are slowed to their group's pace; a lagging member can slow the leader. Group research [R13, R14] motivates coordination, not universal deletion of waiting.

The optional **group-consistent instruction** assigns all group members the original leader's seat-block recommendation. Hearing and acceptance draws remain individual. It changes the message, not group waiting, speed or obedience. Spatial signs always carry their fixed public message and do not acquire personalized group knowledge.

8 V5: route costs and discrete exit choice

People choose only among known exits and links believed open. Dijkstra's algorithm computes the least perceived cost to each eligible exit group. Costs combine walking, visible queues, narrow-door aversion and opposing traffic; wheelchair routes must be accessible. Outward-only links cannot be re-entered inward.

$$\begin{aligned} c_{i,e} &= \text{travel}_{i,e} + \frac{w_Q}{w_T} \frac{Q_e}{C_e} n_e + \kappa_{\text{cf}} \frac{N_{\text{opp},e}}{\text{lanes}_e} + R_e \\ V_{i,g} &= -w_T \frac{T_{i,g}}{30} + w_F F_g + w_S S_{i,g} + 3 w_G G_{i,g} + w_I I_{\text{current}} \end{aligned} \tag{10}$$

n_e=1+1.5 max(0,(2−w_e)/2) for visible gate queues. R_e is zero normally, or an assumed rope-crossing deterrence measured in perceived seconds. Queue and opposing-flow terms are included only where the relevant model can observe them. T_i,g is the shortest accumulated cost. Outdoor routing beyond a building exit has reduced weight; the physical reporting boundary remains the exterior door.

F is familiarity with the main foyer; S is the local fraction choosing g; G is the fraction of accepted manager, LED and preassigned messages that recommend g. Agreeing messages make G=1; contradictory messages divide it. Six behavior types have different weights; those weights remain Mateus's.

$$\begin{aligned} \text{choice}_i &= \arg\max_g (V_{i,g} + \varepsilon_{i,g}), \quad \varepsilon \sim \text{Gumbel}(0,1) \\ P(g) &= \frac{\exp(V_{i,g})}{\sum_{h \in \text{known exits}} \exp(V_{i,h})} \end{aligned} \tag{11}$$

The logit expression follows the standard random-utility derivation [R3]. In the default persistent-taste mode, ε is drawn once per person and exit: it is **not** a fresh independent choice probability at every junction. Group following, knowledge restrictions and commitment further constrain choices. Accepting an instruction is therefore not identical to physically following it through the final exit; both are measured separately.

Initial balancing and occasional reconsideration are supported by route-choice experiments [R7]. They do not imply perfect optimization or that every visible alternative should trigger a switch. Routine re-plans retain a path unless the improvement exceeds the inherited 8 s / 15% path threshold.

9 V6: spatial movement, Model A

Model A resolves each person along a row or lane. A free desired speed is multiplied by terrain, environment, opposing-flow friction and group constraints. Following a leader limits speed by available gap; acceleration relaxes toward the limit while braking prevents overlap.

$$\begin{aligned} v_{\text{des}} &= \frac{v_{0,i} f_{\text{terrain}} f_{\text{env}}}{1 + c_{\text{cf}} \rho_{\text{opp}}} \\ v_{\text{lim}} &= \min\left[v_{\text{des}}, \max\left(0, \frac{h - r_i - r_{\text{leader}}}{H}\right)\right] \\ v' &= \min\left[v_{\text{lim}}, v + \min\left(1, \frac{\Delta t}{\tau_a}\right)(v_{\text{lim}} - v)\right], \quad s' = s + v'\Delta t \end{aligned} \tag{12}$$

h is centre-to-centre longitudinal separation in metres; r are body radii, H=0.7 s and acceleration relaxation τ_a=0.5 s. Default Δt=0.1 s. Lane headway models are motivated by measured speed–spacing relations [R4]; exact lane allocation and parameters are model assumptions.

Rows cap walking at 0.45 m/s and squeezing past seated/preparing people at 0.2 m/s. Original occupants block their row positions until they enter it. Joining a lane requires a physical gap, an open link, gate credit when relevant, and available occupancy. Queues at junctions use a randomized zipper rule. Counterflow splits available lanes; inherited squeeze/swap rules relieve prolonged head-on blockage.

The inherited optional Weidmann-type speed law (historical source linked in [R4]) is:

$$v(\rho) = v_0 \left[1 - \exp\left\{-1.913\left(\frac{1}{\rho} - \frac{1}{5.4}\right)\right\}\right] \tag{13}$$

It remains a sensitivity, not a replacement for the default headway model. A's default non-row occupancy cap is 4.5 persons/m^2. Thus density is constrained by construction; the model cannot predict contact force, crush injury or all two-dimensional arching effects.

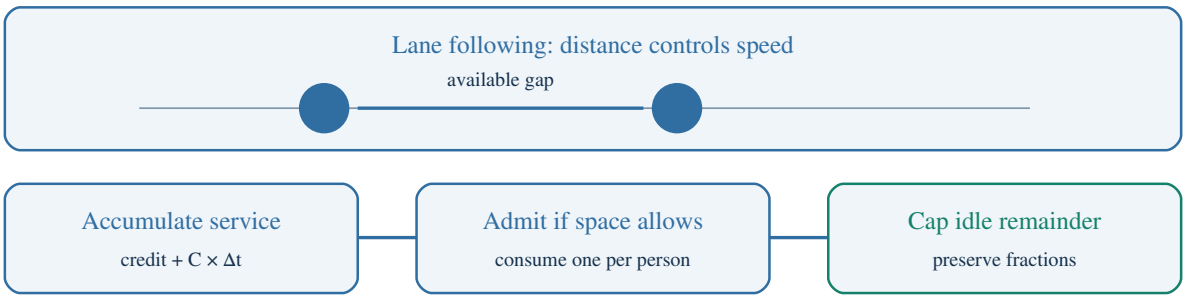


Figure 5: Headway-limited movement and conservative gate service are separate constraints.

The exploratory **vacated-row guidance** test allows accepted staff advice to use existing, fully vacated, visible seatways without opposing walkers. It adds no link, width or speed. Theatre drills observed seatway use [R6], but transferring that behavior to full-house Bailey is uncertain. This A-only route-assumption sensitivity is excluded from the primary unchanged-model intervention winner.

10 V7: managers, pre-announcement and LEDs

Managers activate after a lognormal delay (median 10 s, σ=0.5). In earshot, a new message is heard with probability $1-\exp(-0.35)$ each second. Base acceptance is 0.8. Hearing can create awareness and shorten remaining response through equation (9), teaches an exit and communicates known closures. It does not force the chosen exit.

$$\begin{aligned} p_{\text{hear}}(\Delta t) &= 1 - \exp(-h\Delta t) \\ \text{accepted}_{\text{CM}} &= \mathbf{1}\{u_{\text{CM}} < p_{\text{CM}}\,\text{trust}\} \\ G_{i,g} &= \frac{\text{accepted messages recommending } g}{\text{all accepted messages}} \end{aligned} \tag{14}$$

Legacy managers use same-floor Euclidean reach and one heard-version counter per person; voices can pass through walls. Adaptive managers also require an acoustic room/path check and maintain a version per manager/person pair. The inherited acoustic approximation permits a nearby open-door path within 1.4 times the 12 m voice radius. It is not measured speech intelligibility in Bailey.

Pre-announcement reaches 0.6 of attendees; 0.7 of those accept their seat-block instruction. The product 0.42 is a prior information/adoption probability, not guaranteed final exit usage. Abstract LEDs activate at 3 s, with detection 0.9 and acceptance 0.7; they otherwise know each person's original seat block. In a full-scale station trial, coordinated active signs and non-contradictory voice messages redirected 66% [R8]. **That result does not identify these two separate Bailey probabilities.** Their metadata is now correctly marked assumption.

For spatial signs the continuation tests a visibility catchment [R9]: same floor, maximum range, sign-facing cone and an unblocked sight segment through drawn walls/columns/panels. Each sign carries one fixed exit message. People get one notice/acceptance opportunity when first eligible, using their original uniform draws; repeated checks do not create unlimited chances to notice. This is 2-D geometry: sign height, heads, luminance, smoke optics and installation feasibility remain uncalibrated.

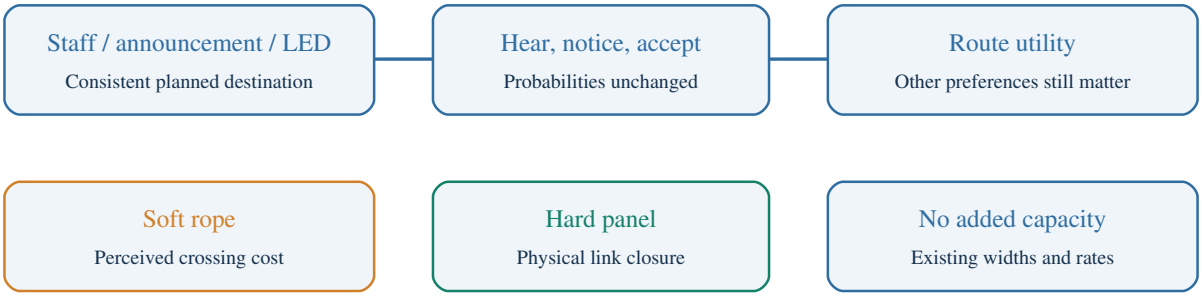


Figure 6: Guidance changes information and preferences; hard panels change traversability.

11 Adaptive individuals and zone-level managers

Mateus's adaptive individual state includes urgency, trust, remembered queues, information age, expected delay and destination commitment. It is enabled independently from adaptive managers. Observations and memories are combined without adding the same queue twice:

$$\begin{aligned} f_{ig} &= \exp\left(-\frac{\text{age}_{ig}}{60}\right), \qquad & B'_{ig} &= f_{ig} B_{ig} \\ T_{ig} &= \widehat{T}_{ig} + \max(0, B'_{ig} - O_{ig}), \quad & \text{confidence} &= \max(c_0, f_{ig}) \end{aligned} \tag{15}$$

O is currently observed delay, B stored delay, and c_0 is 0.8 for foyer/familiar exits or 0.5 otherwise. Seeing the destination replaces its remembered wait. Confidence penalizes uncertain routes. Urgency increases the time weight and reduces the familiarity weight; social susceptibility and cooperation modify social/guidance terms.

$$\begin{aligned} V_{ig}(t) = &-w_T(1 + 0.5u_i)\frac{T_{ig}}{30} - 0.5(1 - \text{confidence}) \\ &+ w_F(1 - 0.5u_i)F_g + w_S s_i S_{ig} + 3w_G(0.75 + 0.5\,\text{coop}_i)G_{ig} \\ &+ w_I I_{\text{current}} - w_I I_{\text{recently abandoned}} \end{aligned} \tag{16}$$

Wait exceeding max(12 s, 1.5×expected), or a surprising queue, triggers review. Cooldown is 6 s; minimum commitment is 10(1−u/2) s. Closures and new instructions can override commitment. Trust falls after conflict/excess delay and can recover; exact gains and losses are assumptions. These mechanisms are retained, not fitted to 180.

Zone managers observe nearby queues after 2 s; radio succeeds with probability 0.9 after 3 s. Every 5 s, they compare travel plus known waits plus anticipated assigned demand/capacity. Switching normally requires an advantage above 20 s and a 20 s hold. They account for pending redirected demand and avoid chains of contradictory diversions. Unknown congestion can still look like zero congestion. [R7, R10] support comparing time and stabilizing guidance; neither validates perfect foresight.

Responsibility counts assigned zones; reach counts people who can hear now; influence counts people still following that manager. Workload is an analytical index relative to 80 people, not a proven staffing limit. Coverage layouts test 6, 10, 12 and 18 staff near seated audiences and existing decision areas. Additional posts receive draws from the same activation distribution. Staff bodies and their own evacuation are absent from the inherited model; the 1,329-person target is the audience, including its three wheelchair users. Extra staffing is an explicit resource and abstraction limit.

12 Adaptive state and staff scoring in detail

Every simulated second, an aware person's urgency relaxes toward an event-dependent target. All quantities below are dimensionless except density ρ in persons/m^2 and accumulated wait W in seconds:

$$\begin{aligned} u^* &= \min\left[1, b + 0.2I_{\text{instructed}} + 0.3I_{\text{closure}} + 0.15\min\left(1, \frac{\rho}{3}\right) + 0.15\min\left(1, \frac{W}{60}\right)\right] \\ u' &= u + (u^* - u)(1 - \exp(-1/10)) \end{aligned} \tag{16a}$$

b=0.2 for the primary tone alarm (0.35 in the separate voice-alarm case). A new staff instruction raises urgency to at least 0.5. A contradicted recent instruction costs 0.125 trust and 3 s hesitation; an agreeing message restores 0.05, capped at 1.2. Excess delay above max(30, 2×expected+20) s costs 0.25 once per decision; trust is bounded below by 0.2. These are inherited behavioral hypotheses, not measured response to a Bailey emergency.

Zone-level managers evaluate the mean estimated route time across participating posts and add anticipated redirected demand:

$$E_g = \frac{\sum_m w_m \left(\text{walk}_{mg} + \sum_{k\in\text{route}} \text{knownWait}_{mk}\right)}{\sum_m w_m} + \frac{\text{forecastLoad}_g}{\max(0.5, C_g)} \tag{16b}$$

w_m is the number expected to move from post m plus one, avoiding a zero denominator. Forecast load includes accepted redirects announced by peers and the candidate's expected movers at each new destination. Current queues already contain their current load, so the controller avoids adding that load again. Cg is the destination's summed exterior gate capacity. Predicted congestion is information, not a change to a physical rate. Closures override the normal 20 s improvement and hold conditions.

The separate staff-workload indicator is:

$$\text{workload} = \frac{H}{80} + 0.5K + 0.5\frac{\max(0, \rho_{\text{local}} - 2)}{2} \tag{16c}$$

H is the current audience in hearing reach; K is the number of watched critical bottlenecks (wait above 30 s). Responsibility R counts still-inside people from assigned zones, and influence I counts people whose current exit agrees with the manager's accepted instruction. Workload above one is labeled overload for comparison only; it is not a validated staffing or safety threshold. Equations (15)–(16c), their observation delays and random hearing jointly define the operational mechanism. Research [R7, R10] motivates comparing information and stable guidance; the exact coefficients are assumptions.

13 The 04:20 chained-FCFS partial controller

The distinguishing 04:20 feature is **partial redirection in A**. Managers are assumed to know everyone's current position, intended route and queues throughout the building, while instructions still require actual earshot. This is explicitly stronger information than local/radio zone management. B does not implement this controller.

For each route, the predictor builds bottleneck stops and walking segments. Nearby parallel doors are pooled; narrow ungated walkways also become estimated resources. Initially seated people contribute expected releases. Arrivals are propagated through sequential queues, with three rebuilding passes to account for downstream congestion.

$$
\begin{aligned}
d_{j,k} &= \max(a_{j,k}, d_{j-1,k}) + \frac{w_j}{C_k} \\
a_{j,k+1} &= d_{j,k} + \text{walk}_{j,k\to k+1} \\
\text{finish}_i &= \text{final departure} + \text{remaining walk}
\end{aligned}
\tag{17}
$$

w_j is the number of people represented by a candidate, including followers moving with a leader. The candidate keeps its place where already queued, avoids counting itself twice, and joins new queues at its arrival time. Saturated gate throughput observations update the nominal predictor rate; they do not change physical gate capacity.

Candidates are examined latest predicted finish first. Move them only when another route improves their own estimated finish by more than a margin, stays within the detour bound and passes counterflow/U-turn checks. After an accepted instruction, remove the old load and add the new load before evaluating another candidate.

$$
\begin{aligned}
\text{move } i \text{ only if} \quad & \text{finish}_i(\text{stay}) - \min_g \text{finish}_i(g) > \text{margin} \\
n^* &= \frac{Q_A/C_A - \text{extraWalk}}{1/C_A + 1/C_B}
\end{aligned}
\tag{18}
$$

The second expression illustrates balancing two queues; clip n* to feasible nonnegative demand. It is not a guarantee for a coupled building. Mateus's recorded problem was optimistic prediction of stage routes before opposing aisle flows developed. New tests preserve the FCFS predictor and restrict new stage diversions or delay the controller's first review. Initial stage users and all exits remain available; these are operational safeguards, not capacity changes.

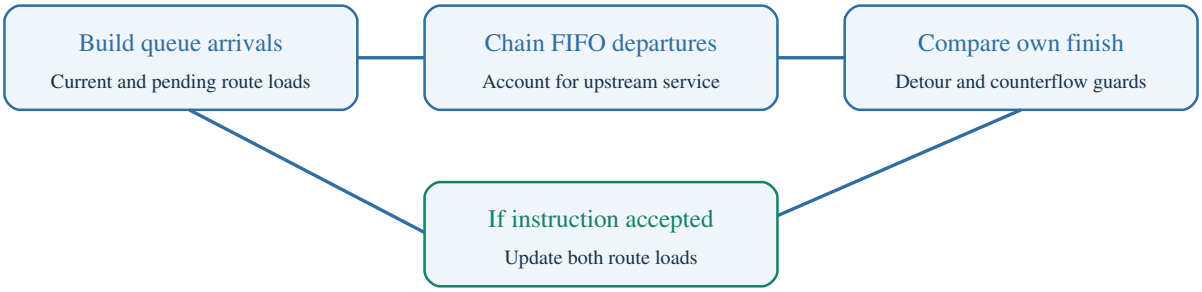


Figure 7: Predictions are updated after each accepted diversion, rather than moving an entire zone blindly.

14 Barriers, visibility and research transfer

Table 4: Barriers, visibility and research transfer.

Device	Representation here	What is not inferred
Hard corridor panel	Closes an existing slot link; opaque to spatial signs	No throughput increase or crush-pressure benefit
Low rope/belt	Route remains physically open; adds perceived crossing cost	No measured Bailey compliance or forced one-way lane

Continued on next page

Table 4 continued

Device	Representation here	What is not inferred
Mesh rail	Same closure as a panel in the present movement model	Optical transparency is not separately calibrated
Shunt posts/bollards	Literature comparison only	Local obstacle-flow gains cannot be pasted onto a graph capacity
S-shaped queue rails	Literature comparison only	A queue-control device is not necessarily an egress accelerator

Wang et al. [R11] reported different effects for diversion posts and safety barriers in their experimental setting. Feliciani et al. [R12] found obstacle effects depend on competition and configuration. Chen et al. [R17] tested S-shaped guardrails with 97 volunteers: time increased even though peak regional density decreased. These results support testing location and device mechanism, not importing an advertised percentage improvement into Bailey.

The rope deterrence sweep uses 5, 15 and 30 perceived seconds. These are transparent sensitivity values, not fitted observations. A rope costs nothing to actual travel time once crossed and never blocks an accessible route. Hard panels can redirect, worsen a queue or obstruct an accessible path; every run must still clear all 1,329 people.

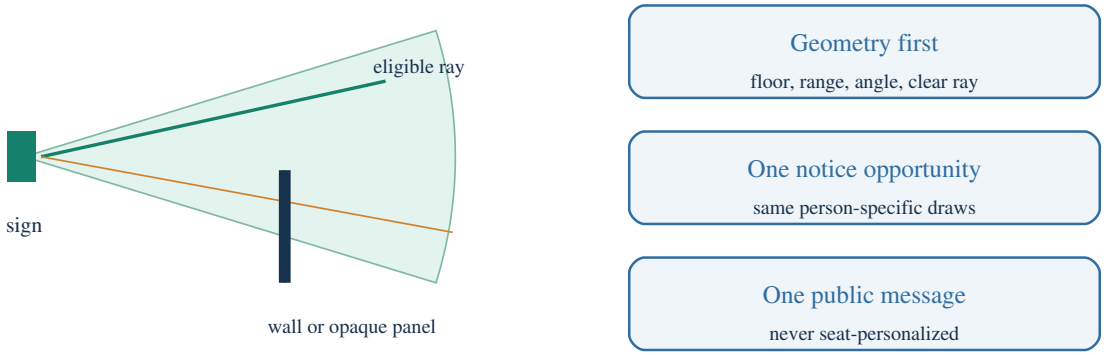


Figure 8: A sign may be close yet unreadable from behind a wall or outside its facing cone.

The Japanese theatre study [R6] had announced drills, 398–540 people on the main floor, symmetric exits and good exit visibility. Its observations cannot establish Bailey visibility, full-house behavior or a mirrored optimal plan. Cornell's current public technical page lists a different seating figure [R15]; the study deliberately retains the **problem's 1,329-person specification** rather than substituting it.

15 Which assumptions can lengthen or shorten time?

Table 5: Which assumptions can lengthen or shorten time?.

Assumption	Likely influence; how treated
Row and aisle headway, effective widths	Strong movement bottlenecks; kept fixed
Group cohesion and waiting	Can prolong the tail; retained, with consistent-message test
Familiar main foyer, incomplete knowledge	Can concentrate demand; addressed through guidance

Continued on next page

Table 5 continued

Assumption	Likely influence; how treated
Other seatways forbidden	Can prevent lateral alternatives; separately labeled A-only sensitivity
Pre-service credit clipping	Artificial service loss; isolated optional numerical correction
Same-floor voices through walls	Optimistic reach in legacy mode; existing adaptive comparison retained
Seat-specific abstract LEDs	Optimistic information access; spatial sign comparisons added
Stage-route forecasts	May be optimistic before queues form; partial-controller safeguards tested
Network contraction/storage relief	Can underestimate spatial delay; retained, limits B's interpretation
Perfectly paired floor plan	Can hide real asymmetry; paired and asymmetric plans compared
Assistance parameters	Affects street safety, not simply building-door clearance; retained

Default population mean speed is 1.2 m/s before age adjustment, elderly share 25%, elderly speed factor 0.78; wheelchair speed is 0.8 m/s. Assistance has two teams, earliest availability 60 s and lognormal descent median 60 s, σ=0.4. These assumptions are not adjusted to make the target pass.

No optimization proves a global optimum. The search compares whole-zone patterns, 44 seat blocks, barriers, placements, communication modes and safeguards, then uses separate audiences for confirmation. Differences between A and B include spatial resolution, queue implementation, group movement and random runtime decisions; they are not a clean estimate of "behavior cost."

Model results are comparative research predictions. There is no calibrated fire/smoke/tenability model, injury model or validated Bailey evacuation trial here. A low mean neither certifies every audience clears by 180 nor determines a safe real installation.

16 Testing, statistics and reproducibility

One seed defines an audience. Interventions use common random numbers, reducing noise in paired differences. Search/training seeds must not be reused as independent confirmation evidence. Raw configurations, seeds, source hashes and per-run summaries are saved; optimization also saves every evaluated plan.

$$\begin{aligned} s^2 &= \frac{\sum_r (T_r - \overline{T})^2}{R-1}, & \mathrm{CI}_{\mathrm{mean}} &= \overline{T} \pm t_{0.975,R-1}\frac{s}{\sqrt{R}} \\ D_r &= T_{\mathrm{candidate},r} - T_{\mathrm{reference},r}, & \mathrm{CI}_{\mathrm{difference}} &= \overline{D} \pm t_{0.975,R-1}\frac{s_D}{\sqrt{R}} \end{aligned} \tag{19}$$

Negative D means faster. The interval quantifies audience sampling variability conditional on this model, not uncertainty about geometry or behavioral calibration. Searching many configurations biases training winners; independent confirmation is essential. A mean below 180 with an interval straddling 180 is weak evidence of meeting the mean target.

$$\mathrm{CI}_{\mathrm{Wilson}} = \frac{\widehat{p} + \frac{z^2}{2R} \pm z\sqrt{\frac{\widehat{p}(1-\widehat{p})}{R} + \frac{z^2}{4R^2}}}{1 + \frac{z^2}{R}} \tag{20}$$

Here z=1.959964 and $\widehat{p}$ is the fraction of complete runs at or below 180. Unlike a normal approximation, Wilson intervals remain informative when zero or all trials pass [R16]. Unresolved runs count as failures;

completion-conditioned statistics are explicitly labeled and never used as an unconditional successful mean.

Density is measured as occupants/area. Local A density uses a roughly 2 m window; B uses edge density. Regional density is comparable in geography but not identical in resolution. Exposure integrates excess density, $\Sigma_i \int \max(0,\rho_i - \rho\text{ref})\text{dt}$; the reference 2 persons/m^2 is an analytical comparison level, not a safety threshold. Peak density and exposure accompany time so a faster, more crowded option is visible.

The exact 04:20 checkout initially passed 124 tests. New tests cover conservative saturated service, idle credit, responder reductions, hard/soft barrier semantics, identical sampled audiences, sign catchments and unresolved bounds. Final counts, fresh holdout comparisons and browser/PDF validation are recorded in the companion report. Reproduction commands and selected full JSON configurations are stored with the study.

Implementation references: `Params.ts`, `Population.ts`, `Behavior.ts`, `Routing.ts`, `Movement.ts`, `QueueSystem.ts`, `NetworkSimulation.ts`, `PhysicalModel.ts`, `StaticFlow.ts`, `TimeExpanded.ts`. These code files, rather than a borrowed paper equation, define the exact numerical model described here.

17 Calibration evidence: could A reasonably be faster?

A network queue model is a legitimate modelling family, but Mateus's B is not independently validated for Bailey. It keeps individual delays, choices and slowest-member group pace while aggregating row travel and aisle motion. It omits A's seated-person squeeze events, actual gaps, lane-order constraints and spatial cohesion stops. Its default density slowdown is off; storage relief after 30 s is an additional optimistic approximation. A's extra detail can also introduce conservative or poorly calibrated restrictions. Neither model is ground truth.

Two existing parameters merit sensitivity tests without replacing the model's equations:

Table 6: Calibration evidence: could A reasonably be faster?.

Parameter	Inherited	Values tested	Evidence and limit
Seat-row speed cap	0.45 m/s	0.60 m/s	Lecture-theatre validation questioned a universal 0.45 cap [R18]; 0.60 is a modelling alternative, not a Bailey measurement
Net time headway H	0.70 s	0.50, 0.60, 1.06 s	Johansson motivates a lower value [R19]; single-file experiments also support larger spacing slopes [R4]

With the unchanged mean body diameter d=0.44 m, the nominal single-lane rate is:

$$J(H) = \frac{1.2}{0.44 + 1.2H}$$
$$J(0.70) = 0.938, \quad J(0.60) = 1.034, \quad J(0.50) = 1.154 \quad \text{persons/s} \tag{21}$$

Reducing H therefore changes assumed physical throughput, even with identical widths. It is a calibration hypothesis, not a guidance intervention. Seyfried's measured relation d=0.36+1.06v applies over 0.1–1.0 m/s [R4]; Johansson's net headway uses a density-based distance approximation. These are related mechanisms with different experimental definitions. Keeping the original body-size distribution while changing H does not reproduce either paper's complete fitted model.

The comparison retains full occupancy, wheelchair users, response, group waiting, compliance and geometry. It varies row speed and H independently and together, and also tests slower movement to avoid one-sided selection. No new parameter is promoted to the main scenario merely because it approaches 180. Local seatway trajectories, saturated aisle flow and group separation data would be needed for Bailey calibration. The companion report shows the sensitivity results and B's existing density/storage options.

18 Numerical verification beyond the 04:20 model

Three new A corrections are independently switchable under `numerics`; absent flags retain the original implementation. They change integration or route validity, not the audience, widths, physical rates or response draws. Full-house effects are measured separately in **SIMPLIFICATION_AUDIT.pdf**.

An admitted transition already carries overshoot into the next link. The old velocity retained only displacement on the old link, causing an artificial slowdown at an empty boundary. `preserveTransitVelocity` includes the admitted carry in the stored step velocity. Carry is limited to the next link; at most one transition per step remains, so very short links still need a time-step check.

$$\begin{aligned} v_{\text{step}} &= \frac{\text{old-link displacement} + \text{admitted carry}}{\Delta t} \\ h_k &= s_{\text{leader},k} - s_{i,k} \\ J_{\text{legacy}} &\simeq \frac{v}{d + v(H - \Delta t)}, \qquad J_{\text{synchronous}} = \frac{v}{d + vH} \end{aligned} \tag{22}$$

`synchronousHeadway` evaluates equation (12)'s gap from start-of-step positions. The inherited leader-first update uses the leader's new position, giving an optimistic discrete-time flow. In a 1.2 m/s paced platoon with followers able to walk at 1.3 m/s, original flow was 1.0345 people/s at Δt=0.1, versus the intended 0.9375. Corrected flow matched 0.9375 at 0.1, 0.05 and 0.025 s. These synthetic tests verify an equation; they do not calibrate Bailey.

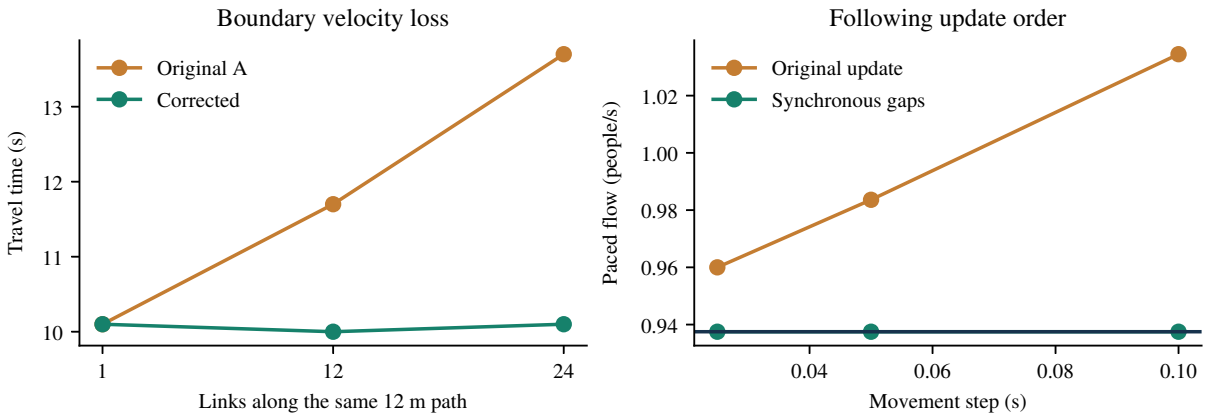


Figure 9: Both conservative and optimistic artifacts matter; correcting them is not guaranteed to shorten evacuation.

`replanMissingPath` invalidates a path that does not contain the current node before applying route inertia. Otherwise an exhausted indoor path can appear to cost zero and delay its street-level continuation after an exterior crossing. This can block people behind the door. A separate regression test reproduces the failure and its correction.

The extension also tests removing acceleration relaxation, seated squeezing, randomized merges or spatial group lag stops one at a time. These are **structural simplifications**, separately named under `simplifica`

`tion`; they are not numerical fixes or guidance achievements. In particular, deleting an acceleration state does not establish that real people accelerate instantaneously.

19 A-Lite: spatial queues between A and B

`LiteSimulation` is a separate engine using A's original **268 nodes and 384 links**, population, route choice, guidance and reporting boundary. It replaces microscopic movement through a supplied `LiteMovement` implementation. It does not use B's contracted network or its storage-relief rule.

Each person travels their seat-to-row-end distance and subsequent link lengths at their own terrain-adjusted speed, capped by their group's slowest free speed and the unchanged row cap. Readiness waiting and leader destination preference remain; spatial cohesion stops and distance-triggered group breakup are removed. This is a simpler group hypothesis, not identical group motion.

$$\begin{aligned} a_{i,e} &= \text{entry}_{i,e} + \frac{\text{remainingLength}_{i,e}}{v_{i,e}} \\ J_e &= \frac{\text{lanes}_e\, v_{\text{rep},e}}{2r_{\text{mean}} + v_{\text{rep},e} H}, \quad S_e = \max(\text{lanes}_e, \lfloor \rho_{\max} L_e g_e w_e \rfloor) \end{aligned} \tag{23}$$

Travel completion joins a FIFO queue for the existing lane. Different lanes proceed concurrently. Rows join their selected end by predicted arrival time, retain finite storage, and have separate end-service resources. Passage directions share J by current directional demand; door and stair entries also require the unchanged physical gate credit. Capacity is never increased to fit 180.

$$\begin{aligned} C_{e,d} &= J_e \frac{N_{e,d}}{N_{e,+} + N_{e,-}} \\ b^* &= b + C_{e,d}\Delta t, \quad n \le \lfloor b^* \rfloor, \quad b' = \min(1, b^* - n) \\ N'_e &= N_e + \text{admissions}_e - \text{departures}_e, \quad 0 \le N'_e \le S_e \end{aligned} \tag{24}$$

Every person in transit or waiting consumes storage. Opposing ready heads can exchange slots only with available service, an open route and enough link width for their diameters. The transaction preserves occupancy; there is no timed overfilling. Remaining deadlocks count as failures. Coordinates retain actual link location for signs and staff, but do not resolve collisions; reported local exposure uses whole-link density. Regional density retains the shared geographical areas.

The separate **SIMPLIFICATION_AUDIT.pdf** derives these rules, audits every major mechanism, reports ablations, independent comparisons, blocked exits and time-step checks, and identifies lost effects. Finite-storage pedestrian flow is supported as a modelling family by Guo, Huang and Wong (2011); representational queue deadlocks are discussed by Palmer, Harper and Knight (2018). Neither validates this Bailey hybrid. A-Lite rejects falls and separated-group stress, and does not model injury or fire tenability.

20 Original brief and fixed-budget placement study

The original seven-page *Bail(ey)ing Out* PDF requires building departure from emergency declaration (pp. 1–2), at least six managers for a full-house concert (p. 2), and prohibits evacuation by elevator (p. 3). The new comparison addresses Questions 2 and 4 with **exactly six managers**, following the user's narrower resource constraint. It keeps 1,329 people including three wheelchair users, all original geometry, rates, speeds, response and compliance. Earlier 10/18-manager results are historical resource comparisons. The

full page-by-page audit is `CONSTRAINTS.md`; the results PDF contains its concise constraint table and the required executive summary.

Temporary barriers in the semicircular corridor are expressly allowed by Question 4. A hard panel closes only its existing corridor slot; a rope keeps it traversable with the inherited assumed 15 s perceived crossing cost. Paired W0/E0, W1/E1 and W2/E2 panels, asymmetric W0/E2 and W2/E0 panels, W1/E1 ropes and no barriers are compared. No positive capacity factor is attached to an obstacle. Every candidate retains graph reachability, including a stair-free route from each wheelchair seat to an outside assistance point. This is a connectivity check, not an installation approval or a crowd-pressure prediction.

The new floor-strip representation adds a finite polyline on an existing walkable link to the earlier point-sign schema. It retains the same one-time notice and acceptance draws and fixed public exit message. Twelve 2 m strips give each layout exactly **24 m installed length**; location is varied between early approach, decision areas and corridor/exit approach. These are localized directional strips, not a claim of continuously illuminated routes throughout the building. The arrow follows a shortest open route to the named exit at installation time. A static strip does not acquire knowledge of a later obstruction.

$$\begin{aligned} \lambda^* &= \mathrm{clip}_{[0,1]}\left(\frac{(x-a)\cdot(b-a)}{\|b-a\|^2}\right), \qquad p^* = a + \lambda^*(b-a) \\ D(x,S) &= \min_{p\in\mathcal{P}(S)} \{\|x-p\| : \mathrm{floor}(x) = \mathrm{floor}(S),\ [x,p] \text{ unoccluded}\} \\ \mathrm{eligible}(x,S) &= \mathbf{1}\{D(x,S) \le R\}, \qquad R = 12\,\mathrm{m} \end{aligned} \tag{25}$$

For each segment from a to b, the candidate set contains the closest projection plus samples no farther than 0.5 m apart, including endpoints. The minimum over visible candidates allows a partly hidden strip to remain detectable; it is an approximation to continuous visibility. Floor strips are viewable from either side, unlike a wall sign's facing cone. This does not model heads, height, photometry or smoke optics. The 12 m range, 24 m device budget and 2 m segment length are experimental design assumptions. Range and notice sensitivities accompany the main comparison.

Bernardini et al. [R20] compared passive and active wayfinding systems using isolated participants in VR. Continuous low-level cues supported some wayfinding tasks, while changing active instructions could confuse users. This supports testing strip location and message consistency, not importing a measured dense-crowd time reduction. Galea et al. [R8] also show the importance of consistent voice and visual instructions. Here the staff and pre-announcement use a seat-block plan; a public strip may disagree with that plan, and the inherited conflict hesitation remains active. Barrier research [R11, R12, R17] motivates testing both benefit and harm without a capacity bonus.

A and A-Lite call the **same guidance and strip-visibility code**, while retaining their distinct movement equations. A-Lite's simpler queue coordinates can change who sees a strip; this is measured model sensitivity, not a reason to assume the same numerical effect. Placement maps and exact coordinate/configuration tables are in the companion results report. Guidance choreography and sign legibility must be understandable to imperfectly informed occupants; a favorable training result alone is insufficient grounds for a sanguine claim about real evacuation.

21 Reproducing a number in the interface

A study mean is not a single simulation. The historical refined A plan uses six existing managers, hard W1/E1 corridor panels, pre-announcement, abstract LEDs, adaptive managers and conservative gate service. Its published mean is **231.2475 s on 40 specified audiences**, whereas the original default A baseline on those audiences is **268.645 s**. The raw configuration seed 1 is not the batch's first seed; the confirma-

tion generator uses base seed 20261108. Repeating seed 1 repeats one audience rather than estimating the mean.

The study tab now offers both a known individual replay and an exact batch replay, plus configuration downloads. The live view separates the last building crossing from the ongoing simulation clock, which includes outdoor travel and wheelchair assistance. Changing a preset replaces the study settings; changing controls after a run has started requires reset. Saved browser settings persist. The 04:20 research checkout is separately identified because older local checkouts may serve different ports and code.

A separate replay audit checks all published main confirmation runs against the current engine, and browser checks exercise the actual live controls and worker pool. The generic Monte Carlo report now suppresses a finite full-house mean if any audience is unresolved. Verification establishes agreement with the implemented equations and archived records; empirical accuracy still requires suitable Bailey observations.

22 Earlier A-Lite recommendation and question scope

The earlier study selected **A-Lite**, with exactly six managers for the concert, as its comparative planning model. The final A-Lite+ extension in Section 24 supersedes that recommendation; the earlier comparison remains documented here. It retains all 268 nodes and 384 original links, heterogeneous people, response phases, group readiness and pace, bounded storage, capacity-limited doors/stairs, imperfect information, blocked-route discovery, accessible paths and actual staff/sign locations. Its finite queues make conservation and service explicit. Choosing it is a decision about the required mechanisms, not evidence that its shorter prediction is more accurate than A. Freeze that choice before inspecting a new confirmation sample.

The modeled path begins along the narrow **seatway in front of a person's own seats**. This is necessary to reach an aisle; it is not walking across seat backs. Subsequent routing excludes all seat-row links when the optional vacated-row sensitivity is off, as it is in every selected plan. Transitions must share an existing graph node and respect closed links, directions and accessibility. The new passive audit checks actual transitions and every occupied row link during complete simulations, including panels and unexpected closures. It does not merely check the initial shortest path. The geometry's polylines connect exactly at their stored endpoints; whether the reconstruction matches every real obstruction remains a surveying question.

A-Lite moves people using travel clocks and queue admission. Its drawn queue coordinates can move backward along the same link as the queue is arranged. These relocations are **not physical trajectories or permission to cross seats**. They also affect approximate local visibility, an acknowledged limitation. Walkway lines are enabled by default in the A-Lite interface. Animation now interpolates along the actual polyline and through its junctions, rather than drawing a chord across a bend; this display correction does not change the simulator. A resolves longitudinal gaps and acceleration, but neither A nor A-Lite resolves full two-dimensional body contact.

Table 7: Earlier A-Lite recommendation and question scope.

Original question	A-Lite application	Limits and required inputs
Q1: ordinary concert departure	Existing ordinary-departure preset; delayed preparation, habit and no managers	Preset population and belongings assumptions need observations; no emergency target interpretation

Continued on next page

Table 7 continued

Original question	A-Lite application	Limits and required inputs
Q2: concert emergency and unexpected blockage	Full 1,329; six managers; unknown blockage learned on approach	Test multiple closures and information delays; no smoke, incapacitation or contact-force prediction
Q3: university lecture	Existing lecture preset has the same occupancy and no managers, with student/familiarity assumptions	Different input hypothesis, not a technique for making the concert pass; lecture response and luggage remain uncertain
Q4: barriers and directional LEDs	Hard/soft corridor-slot semantics, fixed public strip messages, occlusion, imperfect notice and acceptance	No measured rope-crossing resistance, strip photometry, head occlusion or installation feasibility
Q5: optional resource cost comparison	Installation length, barrier count and staff count can be outputs to a cost calculation	Requires actual quotes, labor and training costs; simulation seconds alone cannot establish cost effectiveness

B supports many of these comparisons but contracts walking space, lacks explicit body gaps and allows storage relief after 30 seconds. It is a useful independent abstraction, not the recommended primary model for testing device locations. A-Lite removes that overflow and keeps the original links, but omits seated-person squeezing, detailed merge negotiation, spatial group reunion, falls and injury. It explicitly rejects its unsupported falls and separated-group scenarios. Those are model limits, not mechanisms silently assumed harmless. No single current model can certify safe real-world clearance or answer a cost question without cost inputs.

The **decision engine** is a sequence: become ready; learn exits and closures; estimate travel and observed queues on eligible routes using Dijkstra; score known exits using time, familiarity, social influence, accepted guidance and inertia; add the person's persistent Gumbel taste; choose a reachable exit and follow its path, with occasional revision. Group followers normally follow their leader. A manager or LED changes knowledge and preference, not physical speed or guaranteed obedience. Static acoustic managers are the main operational comparison; the zone-adaptive and partial FCFS controllers are separately described in Sections 10–11.

These simulations are **network-based individual models**, not solutions that force an optimal network flow. Model C alone uses max flow/min cut and time-expanded flow as optimization benchmarks. Maximizing C's flow does not supply an implementable plan for people with imperfect knowledge and compliance. The operational search changes permitted block instructions, placements and devices, then evaluates those choices with the unchanged behavioral simulator.

To test A-Lite, run the application from `mateus-0420` on port 5180 and open `http://127.0.0.1:5180/?tab=lite`. The **180-second study** tab loads exact configurations, replays known seeds and reproduces complete Monte Carlo batches. A-Lite is also available in the Monte Carlo model selector. Use `npmr unlite----seed1466857514--out/tmp/bailey-lite.json` for a single command-line run with the validation-selected overall plan. Add `--configresults/route-study/selected-config.json` for the combined LED-and-barrier plan. These are two prespecified selection roles, not a choice based on confirmation results. The seed identifies an audience, not a promised clearance time. A JSON export includes its complete configuration.

23 What defensible calibration would establish

None of A, B, A-Lite or A-Lite+ is currently empirically calibrated to Bailey Hall. The problem specifies occupancy, floors, exits and constraints; the reconstruction supplies approximate metric geometry.

Fifty-one of its 384 links are explicitly marked approximate, and the absence of that flag does not establish a surveyed dimension. The public Cornell technical page supplies venue dimensions and equipment, not an evacuation validation dataset [R15]. Its different present seating figure must not replace the assigned full-house problem.

Verification asks whether the implementation obeys its equations. Calibration fits parameters to observations. Validation tests predictions against independent observations. NIST [R21] recommends examining response, movement/navigation, exit use, route availability and flow constraints; ISO 20414 [R22] organizes checks according to the model's represented mechanisms. Our conservation, path, rate, time-step and replay tests are verification evidence. They do not constitute complete ISO testing or empirical venue validation.

Table 8: What defensible calibration would establish.

Evidence needed at Bailey	Parameters or assumptions it can identify
Measured clear widths, seatway lengths, stair geometry and available door leaves	Geometry, effective capacity and accessible connectivity
Anonymized start times by seat block and alarm/message timing	Recognition, preparation, social response and staff effect
Seatway/aisle/stair trajectories and saturated gate counts	Row speed, spacing, free speed, stair factors and flow rates
Time-stamped cumulative exits at each door	Queue growth, route allocation and the full clearance curve
Sign visibility/notice, staff audibility and actual route choices	Information eligibility, adoption and choice weights
Group separation and wheelchair assistance observations	Group delays, cohesion approximation and distinct assistance endpoint

Fit several independent observables, not just the final evacuation time. For example, let y_j be an observed quantity, m_j(theta) the model's corresponding mean and s_j a prespecified uncertainty scale in the same units. A possible calibration objective is:

$$\widehat{\theta} = \operatorname*{argmin}_{\theta \in \Theta} \sum_{j=1}^{M} w_j \left(\frac{m_j(\theta) - y_j}{s_j} \right)^2, \qquad w_j \geq 0,\ s_j > 0. \tag{26}$$

Here theta contains only identifiable uncertain parameters, Theta contains evidence-based admissible ranges, and the weights are fixed before fitting. Repeated time samples are correlated, so weighting each second as an independent measurement would overstate information. Retain independent events or held-out observations for validation and propagate parameter uncertainty through the intervention comparison. Equation (26) is a proposed calibration design, **not a fit performed in this study**. No observed Bailey dataset has been substituted with synthetic simulation output.

The 180-second requirement belongs to the subsequent evaluation of a fixed operational plan; it is not a calibration observation. Lowering headway or response solely until the output passes would change the assumed audience or flow capacity. Research-supported alternatives can be tested as labeled sensitivities, but selecting the fastest alternative does not establish its truth. The report therefore states the actual confirmed mean, its uncertainty, failures and device ablations even when the requirement remains unmet.

24 A-Lite+: the primary model and its implementation boundary

A-Lite+ is the primary model for the final competition study. It extends the existing A-Lite transport and shared population/decision code through optional factories. A, B, C and the original Lite remain reproducible. Sections 1–21 derive their common foundations; Sections 22–23 describe the earlier Lite recommendation and its calibration limitations. This section specifies the changes that distinguish the final primary model. The companion results report identifies its completed experiments separately from earlier Lite results.

The original building graph, 1,329 audience members, three wheelchair users, readiness waiting, heterogeneous speeds and response distributions remain. No person can use another seating row as a through-route: only their own seatway leads into the connected aisle/corridor/stair graph. A temporary barrier may narrow or close an approved corridor slot and displace its bypass centreline; it does not reconstruct the building. The endpoint is still equation (1), not the mean person time or assisted street arrival.

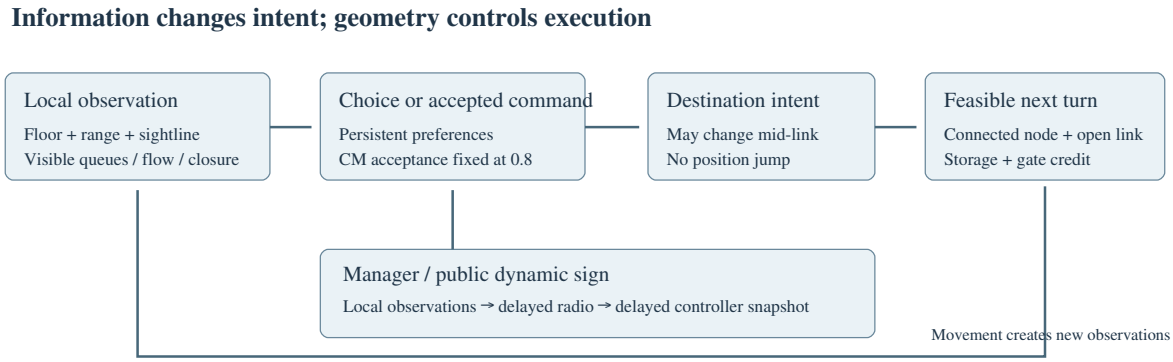


Figure 10: A-Lite+ separates observation, instruction acceptance, destination commitment and physical execution.

24.1 Movement, conservation and the density question

Each original link has finite storage, parallel FIFO lanes, individual remaining distance and conservative service credit. Let n_e be occupancy, A_e admissions and D_e departures during one step. Let d be mean body diameter, H the inherited headway and v_e representative free speed. The same capacity mechanism as Lite applies:

$$\begin{aligned} n_e^{k+1} &= n_e^k + A_e^k - D_e^k, & 0 \le n_e^k \le S_e, \\ S_e &= \max\{m_e, \lfloor \rho_{\max} L_e w_e \gamma_e \rfloor\}, & C_e^{\text{walk}} = \frac{m_e v_e}{d + H v_e}, \\ b_e^* &= b_e^k + C_e^k \Delta t, & b_e^{k+1} = \min\{1, b_e^* - D_e^k\}. \end{aligned} \tag{27}$$

Here m_e is the lane count and γ_e is the inherited geometric area scale (plan length relative to physical link length). The minimum storage of one person per lane can exceed the area-density threshold on a very short link. Both directions share passage storage and service according to their current occupancy; the two ends of a seat row are distinct release resources. The row occupancy limit uses body spacing rather than area density. A downstream admission requires space and, for doors/stairs, original gate credit. Waiting never creates an overflow exemption. A simultaneous opposing-head exchange must conserve occupancy, fit the physical width and consume both required service credits.

A-Lite+ already expresses congestion through finite occupancy, service and FIFO waiting. Applying an additional density-speed curve to the same queued delay would count that constraint twice. It therefore

retains free travel plus capacity-constrained waiting, whose aggregate density-flow behavior is bounded by equation (27). It does not claim a measured Bailey fundamental diagram or predict crowd pressure. Density uncertainty is tested through the storage cap and transferred headway values, separately from operational optimization.

For local group pacing, let $\mathcal{N}_i$ contain the person's unbroken group members on the same link and moving in the same direction within the inherited 4 m cohesion distance. Desired group pace is the minimum free speed in that local set, including the person. Readiness waiting and leader destination following remain. The new rule removes the permanent influence of a remote slow member; it does not add a speed bonus or impose a stop when another member is queued. Bode and colleagues found delayed response and approach among small groups, but no discernible difference immediately before an exit [R23]. Their experiment motivates this distinction; it does not calibrate the 4 m distance.

$$\begin{aligned} v_i^{\text{group}} &= \min_{j \in \{i\} \cup \mathcal{N}_i} v_{0,j}, \\ \ell_i^{k+1} &= \max\{0, \ell_i^k - v_i^k \Delta t\}, \qquad t_i^{\text{ready}} = t_k + \ell_i^k / v_i^k. \end{aligned} \tag{28}$$

Terrain and row limits still apply. Updating the remaining-distance clock when pace changes prevents an artificial position/time jump caused by using a new speed with an old arrival clock. Queue-display positions remain approximate: they may relocate backward within the same link when a queue forms. Visibility uses those approximate positions, so this remains a spatial queue model, not a collision-resolved trajectory model.

A moving follower may change to an actually empty parallel lane on a non-row, non-gated corridor at least 1.8 m wide, when it would otherwise catch its slower lane leader. It does not pass within a narrow lane, reorder a saturated gate queue, cross a wall or skip a graph link. Width and discharge capacities remain unchanged by passing. The width threshold is a stated geometric modelling rule, not a measured overtaking law.

24.2 Local observation, destination commitment and feasible turns

The existing persistent-Gumbel utility and memory rules in equations (10)–(16) remain for unconstrained choices. During movement, people observe visible nearby gates, learn visible closures and compare reachable alternatives. A sight segment must be on the same floor, within range and clear of drawn walls, columns and opaque devices. Known queues can also be communicated by a nearby manager using the inherited delayed local/radio system. Distant true queues are not substituted for missing information.

Queue length is the number of waiting people currently visible to that observer. Apparent discharge is estimated over a five-second continuously visible observation window while a queue exists, bounded above by the known nominal rate. A gap in visibility resets the observation window. Until such a sample exists, nominal capacity is the prior. This is a deliberately simple perception model: there is no calibrated visual counting error distribution.

$$\begin{aligned} \widehat{C}_{i,e} &= \operatorname{clip}\left(\frac{P_e(t) - P_e(t - \Delta_o)}{\Delta_o},\ 0.2,\ C_e \right), \quad \Delta_o \geq 5 \text{ s}, \\ \widehat{W}_{i,e} &= Q_{i,e}^{\text{visible}} / \widehat{C}_{i,e}, \\ \text{review}: &\quad \widehat{T}_{i,g} - \min_h \widehat{T}_{i,h} > \max\{8 \text{ s}, 0.15 \widehat{T}_{i,g}\}. \end{aligned} \tag{29}$$

P_e counts admissions observed at a gate. Walkway and opposing-flow costs retain a coarser observation: when the relevant junction is visible, they use the whole link's directional occupancy. They do not visibility-test each person along that passage, so a long or curved link can convey more information than

an actual observer could count. Gate queues use the separate person-level visibility filter described above. A review retains stable preferences and the existing cooldown; it is not an instruction to choose the shortest route deterministically. A mid-link decision changes the intended destination immediately, but the new path begins at the forward endpoint. The occupied link, longitudinal position and coordinates do not change. The next permitted turn is executed through an adjacent link at a real junction. Logs distinguish a destination change from its physical execution.

Manager hearing, acceptance and action are separate. Hearing still depends on delayed activation, range and the room/open-door acoustic approximation. For each person, the original persistent uniform draw u_i^{CM} determines acceptance. Trust does not alter the primary probability, fixed initially at 0.8:

$$\begin{aligned} H_i &\sim \text{Bernoulli}(1 - e^{-h\Delta t}), \\ A_i^{CM} &= \mathbf{1}\{u_i^{CM} < 0.8\}, \qquad u_i^{CM} \sim U(0,1), \\ g_i^{\text{committed}} &= g_{CM} \quad \text{if heard, accepted and reachable.} \end{aligned} \tag{30}$$

An accepted valid instruction commits the destination and an explicit route. It is no longer merely a utility bonus. Agreeing repeats do not draw new compliance or rebuild an unchanged valid path. The commitment persists until an updated instruction or known obstruction requires a new route. If the destination becomes unreachable, the command is released. Accessible routing takes precedence for wheelchair users. An accepted manager commitment has priority over a conflicting LED message; that conflict is logged. The exact compliance value remains an assumption, rather than a finding from signage experiments.

24.3 Physical signs, strips and delayed public messages

Every additional installation has an ID, fixed position, floor, facing angle, exit message, activation time, range, angle limit, detection probability, acceptance probability and static/dynamic mode. Strips instead follow explicit points within one existing non-row walkway; they do not use a wall sign's facing cone. Existing exit signs are represented at the original exterior/escape-door nodes as baseline information. Their mounting positions are inferred, not surveyed.

At first eligible sight of a message version, a persistent person–installation draw tests detection and a separate draw tests acceptance. Repeated checks do not turn low notice probability into eventual certainty. A noticed message teaches an exit even when not accepted. An accepted LED instruction adopts the destination unless an existing accepted manager command conflicts. Knowledge persists across movement. Updated messages may be read again, using the same installation-specific draws.

$$\begin{aligned} V_{i,s}(t) &= \mathbf{1}\{\text{active, same floor, in range/cone, clear sightline}\}, \\ N_{i,s} &= V_{i,s}\mathbf{1}\{u_{i,s}^N < p_s^N\}, \qquad A_{i,s} = N_{i,s}\mathbf{1}\{u_{i,s}^A < p_s^A\}. \end{aligned} \tag{31}$$

The nominal additional-sign detection value 0.77 is transferred from individual trials in an unfamiliar building: 53 analysed participants aged 18–70, with observed detection 77% head-on and 72% at an oblique approach, versus 39% and 37% for conventional static signs [R24]. Most noticing participants followed. The primary acceptance 0.8, 8 m range and 70-degree half-angle are modelling assumptions, tested separately. Floor-strip transfer is particularly uncertain. The 2017 combined voice/sign result of 66% is not used as a detection parameter [R8]. Baseline static-sign detection is 0.38. Darkness/smoke reductions remain inherited sensitivities; there is no calibrated photometric or head-occlusion model.

A dynamic sign listens to the nearest active same-floor manager. That manager first receives local observations after the inherited two-second delay and may learn remote information by radio after three seconds with 0.9 reliability. The sign controller receives a snapshot after another configured three seconds. It compares nominal route travel plus the waits in that delivered snapshot, using only learned closures. A change

normally requires a 20-second improvement and a 20-second hold. Every viewer sees the same message. A strip reverses its displayed arrow order when the newly indicated route requires the opposite direction. Neither controller reads instantaneous global queues. For a gate within a manager's visibility catchment, the inherited observation uses its full simulated queue count divided by its configured capacity; it does not test visibility of every waiting person or estimate that capacity from a short observation window. This local counting/rate approximation is more informative than an individual's limited visible-queue estimate. A previously reported gate wait persists until a newer observation arrives; missing observations are treated as zero wait, not as evidence that a route is clear. The inherited adaptive-manager demand forecast uses simulated remaining zone counts and peer counts without an additional count-report delay; this is stronger information than the delayed gate observations and is an explicit abstraction. The fixed-plan primary policy does not use adaptive destination allocation. Obstruction experiments introduce one closed link at a time: the inherited manager closure flag refers to the entire configured closure set, so simultaneous closures are not independently discovered by staff in the present implementation.

24.4 Low-profile barriers and explicit bypasses

Only designated corridor slots are eligible. A barrier has physical type, enabled state, length b and angle θ relative to the corridor. Its projected span reduces usable width; the remaining single bypass lies on the other side. The navigation centreline bends around that end, preserving both original junctions and adding the actual modelled detour length. A full-width open rail closes the slot but remains optically transparent. An opaque panel additionally blocks sight; it is retained as a historical comparator.

$$\begin{aligned} w_{\text{bypass}} &= \max\{0, w - |b \sin\theta|\}, \\ m_{\text{bypass}} &= \max\{1, \lfloor w_{\text{bypass}}/w_{\text{lane}} \rfloor\}, \\ w_{\text{bypass}} &= 0 \Rightarrow \text{closed link}; \qquad 0 < w_{\text{bypass}} < 1.1\ \text{m} \Rightarrow \text{reject}. \end{aligned} \tag{32}$$

The 1.1 m threshold is a research screening rule, not a code-compliance finding. Exit/gate/row locations, oversized footprints, duplicate devices and lost seat-to-exit or accessible connectivity are rejected. Every candidate must also be tested under exit obstruction. Belts and open rails do not act as opaque walls, and neither adds a positive throughput multiplier. A-Lite+'s physical barriers have no arbitrary perceived-seconds rope penalty. A low belt may be passed around its end; climbing under or stepping over a closed belt is not modelled. This is a restriction hypothesis, not measured universal obedience.

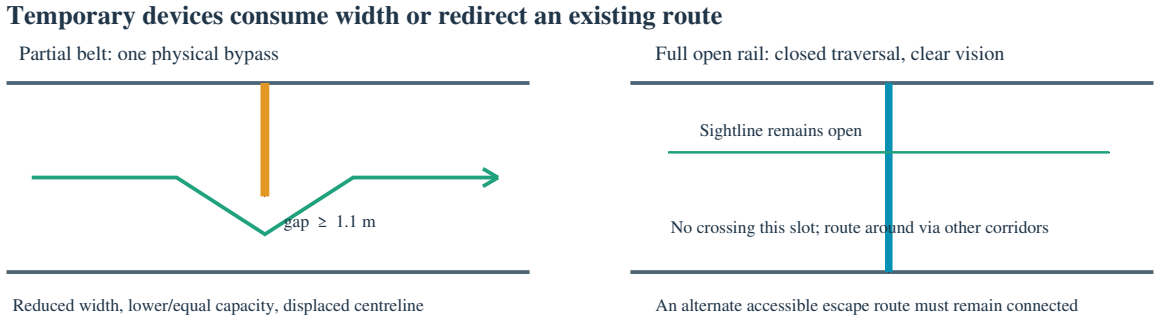


Figure 11: A partial low barrier consumes real width and leaves a displaced bypass; a full open rail closes the slot without becoming a wall to vision.

The device occupies a short graph slot; its reduced-width representation applies over that link. Actual posts, bases and supports must fit within the assessed footprint and leave the stated clear bypass; these hardware details have not been surveyed or separately resolved. A product belt length alone is therefore

insufficient to approve an installation. It does not resolve local body arching or contact forces. Literature demonstrates context-dependent obstacle effects, including slower discharge with lower local density [R11, R12, R17]. A barrier is recommended only if its benefit survives matched uncertainty and obstruction tests. Any physical installation requires review and approval by the University Fire Marshal.

24.5 Calibration freeze, tests and practical use

`src/plus/Calibration.ts` is the compact evidence ledger shown in the UI. The numerical inputs and evidence categories are frozen in `results/plus-study/calibration-freeze.json` before final operational screening. Geometry/occupancy are Bailey problem inputs; quantitative experimental transfers are identified; all remaining coefficients are explicit assumptions. The row-speed, headway, storage, response, group and notice ranges are sensitivity cases, not optimization variables. NIST's stair observations (5,244 journeys, overall mean 0.44 m/s, SD 0.19) measure realized movement including congestion; they must not replace desired free stair speed and then have queue delay added again [R25].

24.6 Frozen evidence ledger

The adopted values below are the primary research configuration, frozen before the placement search. "Bounded" means informed by external evidence and tested over stated alternatives; it does not mean statistically estimated from Bailey observations. Ranges are scenario ranges, not confidence intervals. Other inherited behavioral weights are specified in Sections 6–11 and in the exported configuration.

Table 9: A-Lite+: the primary model and its implementation boundary.

Parameter: adopted value and units	Experimental finding and source	Evidence category, uncertainty and transfer limit
Audience and building: 1329; 527/486/316; 3 wheelchair users (people)	Assigned competition population and supplied plans; 51 link dimensions flagged approximate. CMCM 2026 Bail(ey)ing Out, pp. 1–3ing_Out.pdf)	Bailey problem specification. Range: Fixed. Reconstruction, not surveyed as-built geometry.
Free walking speed: clipped normal, mean 1.2 before age adjustment (m/s)	Single-file experiments relate spacing and speed; they do not identify this mixed audience distribution. Seyfried et al. (2005)	Inherited assumption, research bounded. Range: Mean 1.1–1.3. Elderly share .25 and factor .78 remain assumptions; no Bailey trajectories.
Net time headway: 0.70 (s)	Seyfried reports spacing d=0.36+1.06v over 0.1–1.0 m/s; Johansson studies constant net headway. Definitions differ. Seyfried (2005); Johansson (2009)	Inherited assumption, research bounded. Range: 0.50–1.06. Body diameter .44 m plus H is not the complete fit from either experiment.
Seat-row cap: 0.45 (m/s)	Eight campus lecture-room trials: imposing 0.45 m/s generally overpredicted time; 0.60 was another numerical model, not their measured estimate. Spearpoint & Xiang (2011)	Inherited conservative assumption. Range: 0.45–0.60. Mostly young familiar lecture users; conservative value retained for concert.

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Table 9 continued

Parameter: adopted value and units	Experimental finding and source	Evidence category, uncertainty and transfer limit
Stair movement: individual free speed × inherited stair ratio (m/s)	NIST TN1839 (2015), Table 5: 5244 observed stair journeys, overall mean .44, SD .19 m/s; congested realized speeds must not be substituted for uncongested desired speeds. Kuligowski et al., NIST TN1839	Inherited assumption, externally checked. Range: Ratio ×0.85–1.15. Different stair geometry and occupancy; no Bailey stair calibration.
Door / stair specific flow: 1.3 / 1.0; boundary losses 0.15 / 0.09 (people/(m s); m)	Theatre network trials require aisle and exit constraints; obstacle studies do not establish a universal capacity gain. Spearpoint & Xiang (2011); Wang et al. (2023)	Inherited engineering assumptions. Range: Rates ×0.9–1.1. No measured Bailey saturated flow. Faster-is-slower and luggage factors retained.
Density and spillback: 4.5 maximum; finite shared link storage (people/m^2)	Capacity follows J=v/(d+vH); space is reserved for ALL occupants. No timed overflow. No extra density multiplier on the same queue delay. Seyfried (2005); Guo et al. (2011)	Assumption with conservation verification. Range: 3.5–4.5. Link averages, not contact forces or precise crowd pressure.
Recognition / response / preparation: lognormal medians 6 / 14 / 6; log SD .6 (s)	Unannounced theatre evidence supports response heterogeneity and seat-position dependence. Hopkin, Galea & Deere (2015/2017)	Inherited assumptions; supported distribution family. Range: Response median ×0.8–1.2. Exact medians, social acceleration and spatial coefficients not fitted to Bailey.
Groups: readiness wait ≤90; local pace within existing 4 m on same link (s; m)	108 recorded participants in groups of 3 vs individuals; longer response/approach, no discernible difference immediately before exits. Bode et al. (2015)	Research-motivated structural change; distance assumed. Range: Compare permanent slowest pace. Small artificial groups; experiment does not estimate 4 m. No speed bonus at a gate.
Exit decisions: persistent random utility; 8 s / 15% improvement; 6 s review cooldown (s)	Route-choice experiments support response to travel and queues with relatively infrequent revisions. Liao et al. (2017)	Inherited choice model / assumptions. Range: Controller hold 20–30. Preferences and thresholds not Bailey measured; perceived cost contains only visible/communicated conditions.
LED notice / acceptance: 0.77 / 0.80 per person-installation; range 8 m, cone ±70° (probability; m; degrees)	Controlled unfamiliar-building trials: dynamic-sign detection 77% head-on, 72% oblique versus 39%/37% static; most who noticed followed. Galea et al. (2014), FSS11:1129–1143	Transferred detection; acceptance/geometry assumptions. Range: Detection .50–.90; acceptance .60–.90; range 4–12. Not a dense theatre. Floor-strip transfer especially uncertain; 2017 combined voice/sign 66% is not notice probability.
Existing static sign notice: 0.38 per person-installation (probability)	Static-sign detection near 38% in an unfamiliar building; not repeated independent lottery every second. Xie et al. (2012); Galea et al. (2014)	Transferred experiment. Range: .25–.60. Existing sign positions inferred from exit nodes; exact mounting heights/photometry unavailable.
Crowd Managers: 6; fixed .80 acceptance; 12 m acoustic reach; activation lognormal median10 (people; probability; m; s)	Sign/voice trials support consistent messages; no retrieved Bailey experiment identifies manager compliance .80. Galea et al. (2017)	Count from problem; behavioral assumptions. Range: Acceptance .60–.90; radius 8–12. Hearing differs from acceptance and execution. Local observations delayed 2 s; radio 3 s, reliability .9.

Continued on next page

Table 9 continued

Parameter: adopted value and units	Experimental finding and source	Evidence category, uncertainty and transfer limit
Physical temporary barriers: slot footprint; one bypass ≥1.1 m or alternate-route closure; no gain (m)	Diversion devices can increase or decrease flow; S-shaped rails may lower density while lengthening time. Wang et al. (2023); Chen et al. (2023)	Design assumptions and feasibility screening. Range: No device / partial belt / open rail. 1.1 m is a research screening rule, not code certification. Actual installation needs Fire Marshal approval.

The free-speed distribution has SD 0.25 m/s and is clipped to 0.5–2.0 m/s after the inherited age adjustment. One quarter of the concert audience is classified elderly, with speed factor 0.78. Down/up stair ratios are 0.57 for other ambulant people and 0.51 for elderly people; wheelchair speed is 0.8 m/s. Ambulant body radius has mean 0.22 m and SD 0.02 m, clipped to 0.17–0.30 m; each wheelchair user instead has radius 0.45 m and requires accessible links. Aggregate lane service still uses the representative body diameter, so this is not a separate empirical wheelchair discharge model. These are transferred engineering assumptions, not measurements from the Bailey population. Door flow and stair flow are separate from desired travel speed: a staircase can have a low observed speed because of queues even when its free desired speed is higher. The primary intervention study keeps these values fixed while changing manager posts/messages, pre-announcement and physical installations. The supplementary suite varies them one at a time, also in the slower direction. A favorable sensitivity is evidence of model dependence, not grounds to adopt that parameter after seeing the target result.

Open `http://127.0.0.1:5180/?tab=plus`. Load a catalogue/tested policy, edit signs or strips on the plan, select barrier slots and inspect the resulting bypasses. The engine rejects invalid layouts. A running experiment keeps its existing configuration until reset. The inspector shows known/target exits, visible queues, apparent rates, accepted commands, pending turns and recent decisions. Exported JSON restores exactly the same installations and seeded audience. The comparison panel uses common audiences and reports unresolved runs alongside means and confidence intervals.

Event totals cover the entire run. To bound memory, the general event export retains the first 20,000 entries; each person's inspector separately retains their latest 20 messages and decisions. Accepted messages and the actual destination are displayed separately, since a conflicting sign or an inaccessible instruction need not determine the route ultimately used. Floor-strip validation checks each complete segment inside the union of the walkway's width envelopes, so valid endpoints alone cannot permit cutting across a corridor corner.

Costs are explicitly editable illustrative assumptions, not vendor quotations: capital equals wall-sign units plus strip length plus barrier units plus a dynamic controller if used; staffing is costed per event. Compare total cost across the expected number of events, including local installation and maintenance estimates when available. Additional managers remain a separate resource scenario. Timing, density exposure and closure robustness must be compared before treating two options as operationally equivalent.

Verification covers physical adjacency, conservation, capacity/storage, visibility, instruction execution, public delayed messages, invalid placements, JSON round trips, legacy references and actual browser interactions. Stochastic audience confidence intervals do not absorb uncertainty in transferred parameters or unknown Bailey dimensions. A-Lite+ is research-grounded and verified against its stated rules; without observed Bailey trajectories or evacuation trials it is not empirically validated at this venue [R21, R22]. A sanguine interpretation of a single fast seed is insufficient: reliable choreography depends on clear messages, physical feasibility and legibility.

25 Free speed and distribution shape: a separate, declared sensitivity

The concert default is **1.20 m/s before age adjustment**. The inherited lecture scenario uses **1.35 m/s** and a different audience; it must not be confused with the concert. The user-requested follow-up compares concert means 1.20, 1.35, 1.40 and 1.50 m/s without changing its 25% older share, 0.78 older speed factor, 0.25 m/s pre-age standard deviation, 0.8 m/s wheelchair speed, response, headway, row limits or gate capacities.

Controlled corridor experiments reported individual free speeds around 1.40 m/s for young adults and 1.28 m/s for older adults; older collective movement at low density was slower than the individual pretest [R26]. This supports **testing** 1.40, not identifying Bailey's mixed-audience speed. Natural station observations found substantially lower free speeds and showed that transferring a standard literature value can distort predictions [R27]. "Brisk but normal" is therefore a scenario description, not a venue measurement. No evidence here supports treating every spectator as a fast walker.

The 1.40 m/s clipped-normal alternative was declared in `speed-protocol.json` before any speed-study result. It receives 100 confirmation audiences with the same selected six-manager operational plan. The other means and the right-skew option receive 20 paired sensitivity audiences. Neither the fastest sensitivity nor a single successful seed replaces the declared alternative. The original 1.20 study remains reported separately.

25.1 Exactly what is sampled

Let $Z_i \sim N(0,1)$ be the existing person's speed draw, μ and σ the pre-age mean and standard deviation, and $f_i = 0.78$ for older ambulant people or 1 otherwise. Define $\mathrm{clip}(x) = \min(2.0, \max(0.5, x))$ in m/s. The original sampler clips extremes; it does not resample a truncated distribution. The optional lognormal reuses the same Z_i:

$$\begin{aligned} v_{i,N} &= \mathrm{clip}\{f_i(\mu + \sigma Z_i)\}, \\ s_L^2 &= \ln\left[1 + (\sigma/\mu)^2\right], \\ v_{i,L} &= \mathrm{clip}\{f_i \mu \exp(s_L Z_i - s_L^2/2)\}. \end{aligned} \tag{33}$$

Before clipping, both families have mean $f_i\mu$ and standard deviation $f_i\sigma$. For the lognormal, $E[e^{s_L Z}] = e^{s_L^2/2}$ explains the subtraction in the exponent; its variance is $(f_i\mu)^2(e^{s_L^2} - 1) = (f_i\sigma)^2$. Its median is below its mean. Right skew adds a fast tail while shifting more probability below the mean: it is not a rightward shift of everyone. After clipping and age mixing, empirical moments differ slightly and are reported from the sampled audiences. Wheelchair speed remains separately fixed. No new random draw changes seats, groups, response or compliance.

At $\mu = 1.40$, the pre-clipping older mean is 1.092 m/s and the mixed ambulant mean is approximately $1.40(0.75 + 0.25 \times 0.78) = 1.323$ m/s. These are consequences of the retained age assumption, not a fit to the older experiment. The right-skew family is a distributional sensitivity; the cited studies do not establish it as a superior Bailey fit.

25.2 Why a free-speed change may have a limited effect

The change affects uncongested travel and the representative walkway service rate $J = \mu/(0.44 + 0.70\mu)$. At means 1.20, 1.35, 1.40 and 1.50, this is approximately 0.938, 0.975, 0.986 and 1.007 people/s per lane. Thus 1.20 to 1.40 raises this nominal rate only about 5.2%, despite a 16.7% increase in desired free speed. Door rates, finite storage, row caps, preparation, group waiting and imperfect route allocation remain.

Raising free speed cannot remove those bottlenecks. The gate urgency and faster-is-slower settings are also unchanged.

26 Reproduce the model in the interface

1. From `mateus-0420`, run `npmrundev----host127.0.0.1--port5180`, then open `http://127.0.0.1:5180/?tab=plus`. Use **A-Lite+**, the primary model. Other local servers may be serving a different checkout.
2. In **Tested/catalogue policy**, load `selected-six` for the original concert study or `speed-alternative-1.4` for the declared speed alternative. The selector loads the complete configuration: six staff posts, block instructions, actual signs/strips, barriers and all physical/behavioral inputs. The two speed cases share the operational plan. The **Load confirmed retuned guidance** button loads the completed twelve-manager, 1.4 m/s, 0.0125 s confirmation. The other two planned 100-run fine-step replays were not completed; their confirmation buttons are not shown.
3. Confirm the **Loaded** policy name, seed, time step and staff count above the controls. Inspect the installation map and the current speed/shape in the controls. In the live view click **Reset**, then **Start**. The playback multipliers alter wall-clock playback only. The seed identifies one audience; it is not a mean. Read **Evacuation time (last person out)** in Run results. Continued travel to street level may keep the simulation clock running after everyone has left the building.
4. Open **Compare identical random audiences**, then **Independent confirmation and question results**. **Replay first seed** or the individual-seed menu loads a particular stored audience. Reset and start to compare its displayed clearance time with the recorded value.
5. To reproduce the reported average, click **Prepare 100-run reproduction** on the desired result. This loads its exact policy, sets the recorded seed base, selects **Current editor** and sets 100 replications. Click **Run matched comparison**. The batch derives the same 100 seeds as the command-line study: base 73101 for the original and speed-alternative confirmation, 73403 for the expanded staffing study, or 73503 for the final guidance refinement. A base is not necessarily an individual trial seed. Compare mean, interval, success fraction and unresolved count. The batch can be cancelled; a partial cancelled result is not published as complete.
6. To test a new placement, edit the physical map, export the configuration, choose **Current editor** and compare it with a catalogue reference using common seeds. Invalid bypasses or disconnected accessible paths are rejected. A changed parameter or installation is a new experiment, not a reproduction of the published mean.

The UI exposes both current overrides and the original evidence ledger. A legacy engine cannot silently run an A-Lite+ installation configuration: load a legacy scenario explicitly when comparing A, B, C or original Lite. Browser tests and Node replays check the same exported configuration and seed; neither substitutes for an observed Bailey evacuation.

27 Additional managers, placement and conservative movement

Staffing is an operational resource, separate from the audience and its physical calibration. The expanded study compares 6, 8, 10, 12 and 18 explicit posts, including seat coverage, decision points, bottleneck areas and hybrid layouts. The original six-manager comparison remains available. A requested count must correspond to instantiated posts: A-Lite+ expands the coverage template when more than six are

requested and tests the actual number present. Adding a number to the configuration without adding posts is not a staffing experiment.

For mixed layouts, additional posts are drawn from the existing coverage locations. Seating-zone responsibility is assigned to the nearest same-floor post from the zone's seat centroid. This administrative rule does not grant distant hearing or perfect knowledge. More staff can reach more people earlier, shorten realized response through the existing hearing mechanism and communicate competing directions; extra staff do not multiply a door's capacity or guarantee a benefit.

Research comparing static and moving guides found that guidance mode, speed, relative position and density matter [R28]. The experiment and subsequent subway-platform model do not establish a universal best placement in Bailey. In particular, following a moving leader is different from hearing instructions from a manager who changes post. A-Lite+ tests the latter as a limited operational sensitivity; it does not import a social-force attraction coefficient or a published percentage improvement.

27.1 Explicit patrols on existing corridors

A patrol is a start node, an ordered connected list of existing links, a start time, a walking speed and an endpoint dwell. Its initial post must coincide with the start node: no automatic snapping or teleportation. The search uses 1.0 m/s, starts at 20 or 45 seconds and dwells for 10 seconds at each end. These are declared operational choices, not experimentally estimated Bailey values. Static controls use exactly the same initial posts. Only same-floor, open, non-row, non-gate walkways at least 1.8 m wide are eligible. Stairs and exterior routes are excluded from this patrol comparison.

Candidate construction seeks an eligible node within 4 m of each original coverage post and explicitly relocates the starting post before the run; the matched stationary control uses that same position. No such route exists near the six coverage posts, so those nominal patrol cases remain stationary. The twelve- and eighteen-post layouts each have three actual patrols. This distinction is reported rather than assuming every requested manager can move along a feasible route.

Shared seeds guarantee the same sampled audience, not identical runtime random decisions under every configuration representation. Inherited guidance draws activation times for all available geometry posts before hearing events. An explicit six-post list consumes fewer such draws than the ten-post base geometry, even when its six active positions and activation times match. Its later hearing sequence therefore differs. Compare moving cases with their own explicit stationary controls, and do not interpret that representation-induced sampling difference as a physical placement benefit.

Let s_m be the manager's distance along the current link of model length L_e, n_e its audience occupancy, and t_m^{ready} the later of activation, scheduled start and any dwell expiry. The manager's discrete advance is:

$$\begin{aligned} \Delta s_m &= \mathbf{1}\{t \geq t_m^{ready},\ n_e = 0,\ e \text{ open}\} \min\{v_m \Delta t, L_e - s_m\}, \\ s'_m &= s_m + \Delta s_m, \qquad \boldsymbol{x}'_m = \boldsymbol{p}_e(s'_m / L_e). \end{aligned} \tag{34}$$

The function $\boldsymbol{p}_e$ interpolates the actual walkway polyline. At a junction, only the next connected patrol link can be entered; at either end the manager waits before reversing. If any audience member occupies the link, motion pauses. Movement logs record distance, waiting and arrivals. This is deliberately restrictive: the model does not give a moving manager free passage through a dense queue. It may therefore find little benefit from patrols during peak congestion.

Hearing follows the current manager position. Each second, acoustic distances, local observation sets and peer overlap are refreshed; visible gate information still requires line of sight. Observation, radio and public-sign delivery delays remain. Walking does not refresh a person's persistent acceptance draw,

extend voice range or create a new instruction version merely to obtain another compliance opportunity. The live map shows actual moving posts, while the editor and report show starting posts and purple patrol paths.

As in Mateus's fixed-post abstraction, staff bodies are not included in audience queue storage or the 1,329-person clearance endpoint. Consequently this is an information-coverage sensitivity, not a validated simulation of staff–crowd contact or staff evacuation. The empty-link motion restriction reduces the unsupported claim, but does not remove that limitation. A patrol result is compared with its stationary counterpart and cannot certify an installation or staffing plan.

27.2 Selection and uncertainty

The expanded operational search uses the separately declared 1.40 m/s alternative, eight screening audiences (base 73401), then sixteen validation audiences (73402) for four shortlisted policies. The lowest complete validation mean is frozen before 100 new audiences (73403). The same staffing policy at 1.20 m/s and the six-manager 1.40 control use those same confirmation audiences. This separates resource effects from the change in speed calibration. Finer time steps, three closures and a stationary-post comparison use another twenty audiences (73404). Neither physical speed nor a holdout audience is selected to manufacture success. A further protocol, declared before these confirmation results, always repeats both the six-manager 1.40 policy and the expanded-staff 1.40 policy at the finest 0.0125 s step on their respective 100 confirmation audiences. The original step remains reported; the fastest step is never selected.

Costs scale with the actual staff count. At the illustrative $120 per manager-event, 12 staff cost $720 more per event than six, and 18 cost $1,440 more. Training, placement feasibility, conflicts between messages and obstructed-exit performance remain part of the comparison. A faster mean alone is insufficient to recommend the most expensive configuration.

27.3 Final operational follow-up

The expanded staffing policies initially inherit messages optimized for six managers. Experiments on queue-sensitive route choice and simulation-based guidance motivate testing allocation [R7, R10]; they do not prescribe the search budget or guarantee an improvement. A final bounded follow-up therefore keeps the selected posts, declared 1.4 m/s calibration, LEDs and barriers fixed, and changes only staff/pre-announcement destinations. Eight training audiences compare four coherent west/east side-zone patterns and the inherited plan, then one sweep over the eight non-side seat blocks most represented in the late tail. An improvement must exceed one simulated second on the common training audiences. Three candidates receive twenty independent validation audiences; the lowest complete mean is frozen before 100 new confirmation audiences and a mandatory 0.0125 s replay of the same audiences. Three unknown closures are tested with and without the barriers. The search stops after this protocol regardless of whether 180 seconds is reached. Its configurations, all evaluated plans, sources and outcomes are archived; the UI labels the result **retuned guidance**. Public signs retain their original spatial messages.

27.4 Numerical accuracy of the frozen policies

For an isolated constant-speed traversal of duration τ, checking arrival only on the time grid gives $\lceil \tau/\Delta t \rceil \Delta t$, with rounding error $0 \le \lceil \tau/\Delta t \rceil \Delta t - \tau < \Delta t$. For an uncongested path of m independently checked passages, these travel-rounding errors sum to less than $m\Delta t$. This is not a bound on the complete interacting evacuation: queue order, hearing and route decisions can also change when event times shift.

The expanded-policy twenty-audience means changed by 2.32 s between steps 0.025 and 0.0125 s. Before opening the 100-run fine-step results, a further numerical diagnostic was declared at 0.00625 s on the first twenty matching confirmation audiences of each frozen policy. Only the integration step changes. The report retains the declared 100-run 0.0125 s results, shows paired changes at the smaller step and discusses remaining numerical uncertainty; it never selects whichever step is fastest.

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